S.C.O.P.E.

Smart Cross-camera Observation and Person RE-identification System

COURSE: HS202	INSTITUTE
HUMAN GEOGRAPHY AND SOCIAL NEEDS	INDIAN INSTITUTE OF TECHNOLOGY ROPAR

TA IN CHARGE: SUMAN KUMAR

SIDDHARTH SINGH	2024MMB1422
ASHISH SINGH	2024EEB1406
VANSH GARG	2024MEB1397
LAKSHYA GOYAL	2024MMB1418
SOHAM D. PADOLE	2024EEB1249
LAKSHAY GUPTA	2024EEB1219
VARUN CHOUDHARY	2024MEB1398
SHIVAM GOYAL	2024EEB1388
VINIT KUMAR CHOUBEY	2024MEB1400
KSHITIJ GUPTA	2024EEB1218

Abstract

Universities and colleges often face practical challenges when it comes to maintaining **safety** and security on campus. Even though CCTV cameras and security personnel are present, most monitoring is still **manual**, which makes it difficult to prevent incidents in **real time**. A recent **theft incident** in Chenab Hostel at IIT Ropar clearly highlighted this issue. In this case, an unauthorized individual entered the campus through **impersonation** and was able to move freely inside the premises, eventually stealing multiple laptops.

This incident revealed several **gaps** in the current surveillance system, particularly the lack of real time monitoring, proper **identity tracking**, and timely **alert mechanisms**. Current CCTV systems mainly act as recording tools and are not capable of actively detecting or responding to suspicious activities as they occur.

To address these challenges, this project proposes an **AI based** multiple camera surveillance system that combines **person detection**, **tracking**, and **cross camera identity recognition**. The system is designed to support security personnel by providing real time insights and generating alerts based on **movement patterns** and potential risks.

By improving **situational awareness** and enabling **quicker response**, the system aims to enhance overall **safety**, reduce the chances of such incidents, and make surveillance more **effective** and **trustworthy**

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Introduction

Some basic things which any student expects from their college is **safety** and **security**. A **safe environment** doesn't only increase trust in the institute but it also supports learning and increases productivity. To address security concerns many institutions have adopted **Closed-Circuit Television (CCTV)** systems as a primary surveillance tool. So, CCTV cameras are commonly installed **at hostel entrances, corridors, parking areas, academic blocks** and other sensitive locations. Their basic purpose is to **record activities** and assist investigations after any incident occurs.

However a majority of the existing CCTV systems function as **passive** recording devices instead of **intelligent monitoring systems**. They capture footage continuously, but they can not automatically interpret events or track suspicious movement. We also need systems which can **integrate multiple cameras at once**. The most obvious solution is continuous monitoring by **security personnel**, but this is not only **time consuming** and difficult but it is prone to **human errors** as well. In most of the cases footage is reviewed only **after an incident** has already taken place. So, the presence of cameras alone does not guarantee effective security.

The motivation for this project came from a **major wake-up call** right here on campus. Some students' **laptops got stolen** in broad daylight out of our hostel. There were **CCTV cameras** set up all over the place, so one would think catching the guy or at least **tracking where he went** would be easy but it was not true. The laptops were gone and once the **thief left the hostel** he was not found. Watching the aftermath exposed just how **flawed** our everyday security setup really is. The cameras recorded the footage just fine, but that was it they were just **standalone video feeds**. There was no system connecting them to **track the suspect** seamlessly from the hallway, to the stairs, to the main gate. Instead, **the security guards** had to sit down and **manually** go through **hours of disconnected footage** from a dozen different angles. It took way too long and in situations like this **time is everything**.

It hit us that the problem wasn't a lack of cameras, it was a **lack of brains** in the system. That real-world frustration is exactly what drove this project. We realized that just slapping cameras on walls doesn't work anymore unless there's a **smart system** backing them up that can actually **understand movement**, keep track of who is across different video feeds and give security a **real-time advantage** instead of just a recording of what went wrong.

So, we can conclude **traditional surveillance** systems are **passive** in nature and only **store videos** leaving the most difficult task of **interpreting** to humans. To deal with this we need **intelligent surveillance systems** which can combine the cameras with **AI** to help **automate** this tiresome task and **eliminate human error**.
Instead of simply recording events, such systems can **detect persons automatically**, track movement within and across cameras, recognize repeated appearances, generate alerts, and visualize activity patterns for operators. These requirements are the basic idea behind our project **Smart Cross-camera Observation and Person rE-identification System** which is designed to upgrade existing **CCTV systems** to an **active**, **well connected** and intelligent system.

Defining the Problem

The real issue here is "**surveillance failure.**" This doesn't mean the cameras were broken. It means they perfectly recorded the incident, but the system completely failed to **spot anything suspicious** or help security **stop it in time**. We had footage, but **zero useful outcomes**. Basically, the setup just reacts to crimes after they happen, rather than actually preventing them in the first place.

The main problem with the current system is not the unavailability of cameras but the system not being able to take **instant decisions**. Many institutes and business organizations have installed **CCTV systems** but the occurrence of events like theft and unauthorised moments are still common. This is because most of these systems are only **passive recording devices** and don't warn the authorities **fast enough** in order to prevent such incidents.

Another massive headache is **tracking people.** In busy spots like hostel corridors or campus pathways, everyone is constantly moving. The problem is that our current system treats every camera like a **separate entity** i.e. when a suspect walks out of one frame and into the next, the system completely **forgets who they are**. So, the trail just goes cold. This creates huge **blind spots** during investigations, forcing security to manually go through several different video feeds just to find one person's path.

Things get even worse with **real-time monitoring**. We expect security guards to stare at dozens of screens simultaneously, which is honestly impossible. **Human attention drops** and it's super easy to miss something sketchy happening across a massive campus. Even if a guard does spot something weird, the response is painfully slow because there's no automated system giving them **instant alerts**.

Basically, the core problem is that our current surveillance is **disconnected, passive, and totally relies on manual effort**. It can't track people across cameras or flag **real-time risks**. To fix this, we need to upgrade from basic recording to a **smart, connected system** that continuously tracks movement and helps security act instantly.

Problem Statement

While the cameras continuously produce **video footage** but this information still remains difficult to use in case any **incident** occurs. As these cameras are **disconnected** it is difficult to trace the **path a person took**. Any investigation is highly dependent on **manually reviewing** these footages and this can delay any further action.

As campuses become larger and movements increase, **traditional monitoring methods** are no longer enough. The problem addressed in this project is the lack of an **intelligent system** which can automatically connect visual information, interpret **real time moments** and support security personnel with **instant alerts.**

Enhancing Women Hostel Security Through AI Surveillance and Real-Time Monitoring

INTRODUCTION

Educational institutions together with residential campuses now encounter security problems which include unauthorized entry and theft and assault and suspicious activities. The primary function of **traditional CCTV** systems consists of their role as passive recording devices which do not offer any capacity for real-time threat identification and emergency response. Artificial Intelligence integration with surveillance cameras enables systems to monitor activities through four functions which include intrusion detection and suspicious behavior analysis and person tracking and automated alert generation.

ANI news article

The study investigates the 2024 security breach at the Osmania University Ladies' Engineering Hostel in Hyderabad while assessing how artificial intelligence surveillance systems could have stopped or reduced the actual danger.

NCRB DATA

According to the National Crime Records Bureau (NCRB), crimes against women in India increased significantly over the past decade which shows that surveillance systems and quick response systems should be implemented in sensitive areas like girls' hostels and educational campuses.

The NCRB Crime in India Report 2019 recorded over 405000 crimes against women nationwide.

SOURCE: NCRB Crime in India Report 2019 https://ncrb.gov.in

BACKGROUND OF THE INCIDENT

Security problems on Indian educational campuses which particularly affect women's hostels have developed into a major problem during the last few years. The **NCRB Crime** in India Report 2019 documented more than **4 lakh** incidents of violence against women throughout the country which demonstrates the urgent requirement for improved monitoring and entrance restriction systems. An outsider entered the hostel premises at night during the 2024 Osmania University Ladies Engineering Hostel incident to steal but he ended up threatening a student and escaped before security personnel arrived. Traditional **CCTV systems** exhibit operational limits which the incident demonstrated while demonstrating the necessity of using **AI-based smart surveillance systems.**

CHRONOLOGICAL BREAKDOWN OF EVENTS

1 Late Night Intrusion:

- An outside individual approached the girls' hostel area during the **late hours of the night** when there seemed to be less security and student activity.
- The absence of continuous **perimeter monitoring** enabled the intruder's near-instantaneous movement near the campus.

2 Unauthorized Entry:

- The intruder gained entry into the hostel area because the security system lacked effective access control measures which required **biometric verification** and smart ID authentication for entry.
- Unfortunately, there wasn't any **AI-powered visitor-sensing or intruder-alert system** in the restricted housing society to detect unauthorized entry.

3 Movement Inside Hostel:

- In the hostel, I stepped inside and walked through some **internal hallways** to get to the students' rooms.
- A **lack of intelligent tracking cameras** and active monitoring actually encouraged him to roam freely around.

4 Theft Attempt:

- An intruder sneaked into a room taking advantage of a gateway that was ajar and **attempted stealing a mobile** phone and other stuff inside.
- Because the surveillance system serves as merely a recording medium, no emergency alerting, no immediate warning.

5 Threat Escalation:

- It was said that the intruder **threatened** a student with a **knife** when a student found him in an attempt to escape.
- First, there was an attempt to break in, but it escalated into a **security breach**. Panic and terror crept in among the residents of the hostel.

6 Escape:

- The intruder managed to escape the hostel area after the solitary guard could not **act in time.**

ANALYSIS OF SECURITY FAILURES

A. Lack of Access Control

The hostel lacked advanced access-control mechanisms such as **facial verification, smart gates, and digital identity authentication systems.** The absence of these systems allowed unauthorized people to enter the restricted residential area without being detected or verified.

B. Passive CCTV Infrastructure

The current CCTV system operates as a basic recording system which lacks advanced smart monitoring capabilities. Standard CCTV cameras have the ability to record events for future analysis yet they lack the capability to identify **unusual behavior** and provide immediate security notifications.

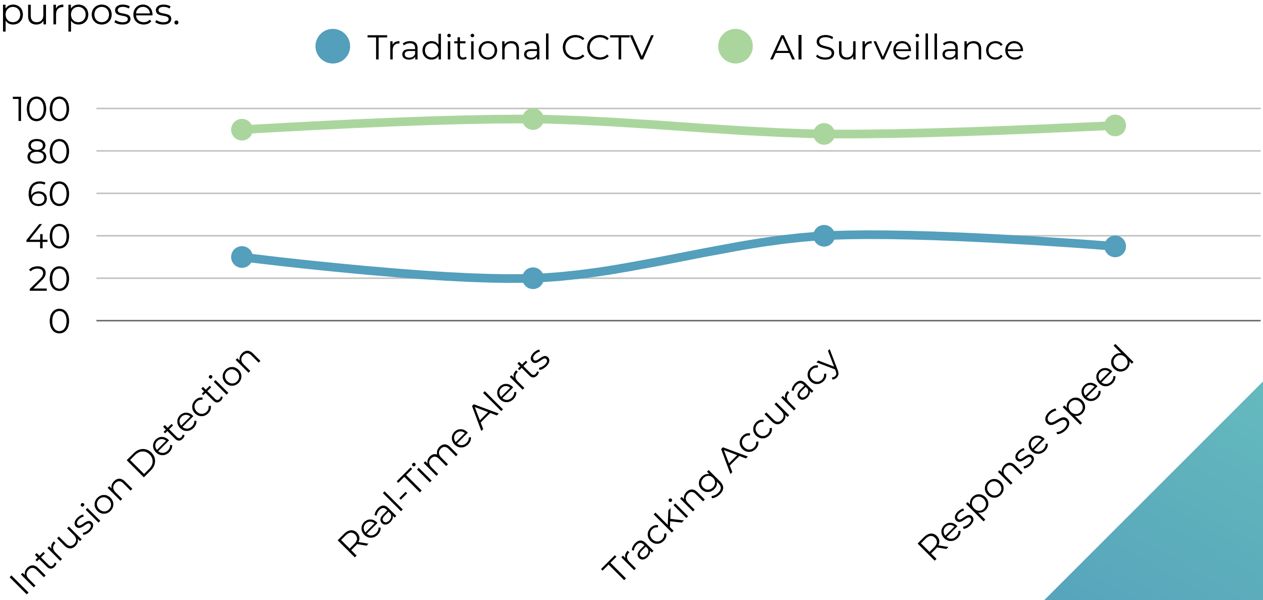
C. Delayed Human Response

Security personnel became aware of the incident only after the situation had already escalated, which caused them to **delay their response** to the situation. The emergency response time decreased because the system lacked both **automated alarms** and **immediate alert systems.**

Osmania University girls dorms

D. Absence of Behavioral Analytics

The security system lacked video surveillance systems which combined **artificial intelligence** with advanced behavioral analysis features that detected loitering and monitored unauthorized access and tracked **potential threats.** The hostel system failed to detect unusual activities which continued until people directly observed them. Studies show that AI-powered surveillance systems achieve **60 to 70 percent faster** response times than traditional CCTV systems which require human monitoring for security purposes.



AI-BASED THEFT PREVENTION FRAMEWORK

PHASE 1
Intrusion Detection

Artificial intelligence-powered cameras could track movement across certain boundaries or visual lines within restricted segments and fairly instantaneously **detect potential unauthorized intrusion** into hostel premises.

PHASE 2
Facial Recognition

Utilizing a base of registered students and authorized staff, the system is **able to verify** faces in order to identify unauthorized visitors for entry into the hostel.

PHASE 3
Loitering Detection

AI analytics can identify people who exhibit suspicious standing behavior or show **unusual movement** patterns near gates and corridors and hostel rooms during late-night hours.

PHASE 4
Multi-Camera Tracking

A networking lab should take real images of its **student's data** with any kind of natural or digital behaviour: chatting online or playing games.

PHASE 5
Automated Alert System

If a sign of suspicious activity is identified, the system could generate **SMS alerts, mobile notifications, alarms, and dashboards** on a real-time basis and supply the data directly to security personnel.

EXPECTED BENEFITS

The expected benifits for using our AI detection system are stated as follows:

Improved Hostel Safety:
Continuous monitoring enhances student security within hostel premises.

Rapid Response:
Real-time alerts allow **faster action** during emergencies.

Evidence Collection:
Recorded footage can assist in investigations and suspect **identification.**

Psychological Deterrence:
AI surveillance can discourage **intruders and criminal behavior.**

Centralized Monitoring:
Multiple cameras can be monitored through a **single security dashboard.**

CHALLENGES AND ETHICAL CONCERNS

Privacy Concerns:
Continuous surveillance may raise concerns regarding **student privacy** and **personal space.**

Data Security:
Recorded footage and user data must be securely stored to **prevent misuse** or **unauthorized access.**

Facial Recognition Ethics:
The use of facial recognition systems may lead to ethical concerns related to **consent and identity tracking.**

Storage Limitations:
Continuous video recording requires **large storage capacity and regular data management.**

False Alarms:
AI systems may occasionally generate incorrect alerts due to **detection errors** or **unusual normal behavior.**

CONCLUSION

The Osmania University hostel intrusion incident demonstrates the limitations of traditional surveillance systems which operate in confined residential areas. Educational institutions and sensitive environments experience reduced thefts and suspicious activities through **AI-powered smart surveillance systems**, which deliver proactive threat detection and real-time response capabilities together with automated security enforcement.

Theft incident at IIT Ropar- The Motivation for S.C.O.P.E.

BACKGROUND OF THE INCIDENT

IIT Ropar educational institutions depend on traditional security systems which include security guards and CCTV cameras for their security needs. The security systems function as **passive systems** which record events but do not perform active surveillance of suspicious behavior or issue notifications during live security events. The methods might seem adequate for regular situations but they break down when criminals execute their plans through **deception and human mistakes.**

A **theft incident** that occurred at **IIT Ropar** exposed several critical vulnerabilities in the existing **surveillance and monitoring framework.** The offender successfully entered the campus, moved across different zones, and repeatedly accessed hostel premises without being detected or questioned properly. The incident shows how manual monitoring systems **fail to work properly** because they need intelligent surveillance systems which can automatically detect suspicious activities.

OBJECTIVE OF THE CASE STUDY

The purpose of this case study is to analyze:

- How the theft was executed
- Weaknesses in the **conventional surveillance system**
- **Human and procedural failures** involved
- The need for AI-based intelligent surveillance systems

TIMELINE OF THE INCIDENT

1. **12:11 PM – Entry into IIT Ropar**
A **45-year-old man** entered the campus claiming to be a Zomato delivery agent. The man arrived at the campus with a black backpack and a **Hero Splendor motorcycle.** The guard allowed entry without identity verification.

2. **12:16 PM – Arrival at Chenab Hostel**
The individual reached **Chenab Hostel** and identified himself as a faculty member. The absence of **ID verification and visitor entry documentation** enabled people to enter the hostel without restrictions.

3. **12:18 PM – 12:40 PM – Theft Execution**
The intruder then went to the third floor looking for open or unguarded doors during low time zones.

Estimated Theft
- **17 laptops stolen**
- Average value: **₹60,000–₹80,000** each
- Total loss: **₹10–13 lakh**

The items were transported in multiple trips using the backpack and motorcycle storage.

4. **~1:00 PM – Exit from Campus**
The individual exited the campus on the same vehicle without any questioning or anomaly detection at the exit gate.

SECURITY FAILURES AND VULNERABILITIES

Human-Based Security Failure

The guards used their complete trust on verbal statements because they did not use any verification methods. The offender used **human psychological techniques** to present himself as a delivery agent and later as faculty. The guards permitted entry because people tend to trust **authority figures** without questioning them.

Lack of Identity Verification

The campus lacked a system that could **monitor student identity** throughout all their locations. The same person entered multiple zones and re-entered the hostel several times without being **recognized as suspicious**.

Temporal Exploitation

The theft happened between **12 PM and 1 PM** because that time had the highest chance of success since most students were either attending classes or eating lunch. At this time, hostel activity remains **low while security personnel focus** on other tasks.

Spatial Vulnerability

The offender specifically targeted the **third floor of the hostel** because of lower footfall and reduced chances of interruption. The suspect had prior knowledge about how people moved through the **campus and how the hostel operated.**

Chenab Hostel entry gate

Passive CCTV Monitoring

The **CCTV cameras** which had been installed at the location operated only as recording equipment. The system lacked both **real-time monitoring capabilities and automated alert systems** which would have detected suspicious activities.

BEHAVIORAL PATTERN OF THE OFFENDER

The **theft was not random but carefully planned.** The offender demonstrated:

- Prior reconnaissance of the campus
- Knowledge of hostel activity timings
- Use of multiple fake identities
- Use of ordinary objects to avoid suspicion
- Confidence in exploiting security loopholes

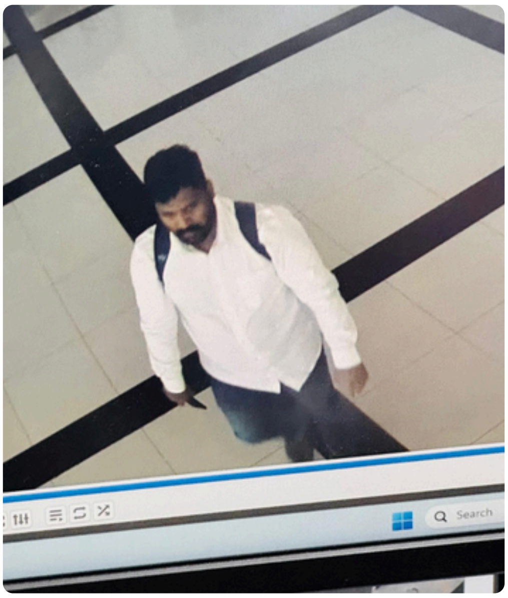
This shows that the failure was not caused by a **single guard's** mistake but by weaknesses in the overall surveillance system.

FINANCIAL IMPACT

The total estimated loss from the theft was around ₹10–13 lakh. Students experienced financial loss together with interruptions to their academic activities because their **laptops and personal data were lost.**

PSYCHOLOGICAL IMPACT

The incident decreased the **security feelings and trustworthiness** of the hostel residents.Students developed increased worries about how well campus security systems protected their safety on the school grounds.



Real image of the theif

INSTITUTIONAL IMPACT

The theft raised serious concerns regarding:

- Existing hostel security protocols
- Visitor verification systems
- Dependence on manual monitoring
- Need for technological upgrades in surveillance

KEY LEARNINGS

This case study demonstrates several important lessons:

- Manual surveillance alone is not reliable for large campuses
- **Human judgment** can be easily manipulated through deception
- **Passive CCTV** systems are reactive rather than preventive
- Real-time monitoring and automated **alerts** are necessary
- Identity tracking across different campus zones is essential

The incident clearly shows the **limitations of conventional surveillance systems** in handling modern security threats.

NEED FOR AN AI-BASED SURVEILLANCE SYSTEM

An **AI-powered intelligent surveillance system** could significantly reduce such risks by:

- **Tracking individuals** across multiple cameras
- Detecting **unusual movement patterns**
- Generating **real-time alerts** for suspicious activities
- Maintaining identity consistency throughout campus
- Assisting guards in faster decision-making

Such systems can enhance security by combining human supervision with automated analysis and **behavioral monitoring.**

CONCLUSION

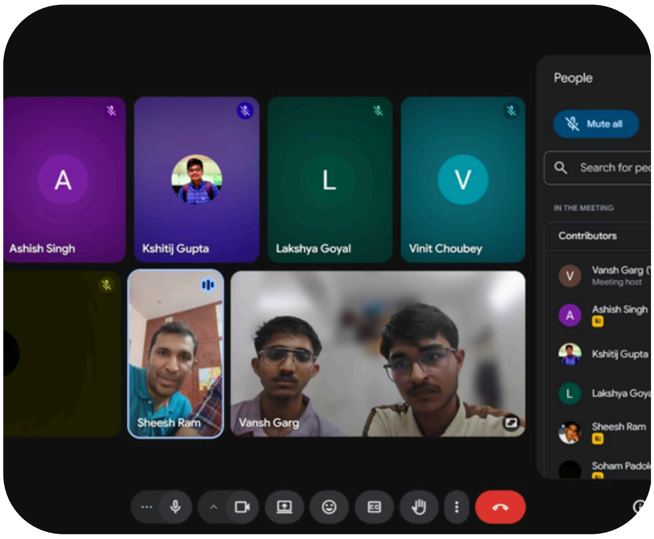
The **IIT Ropar hostel theft incident** shows how traditional security methods fail to protect against theft. The offender successfully exploited human trust, **lack of verification protocols**, and passive surveillance systems to commit large-scale theft without detection. The situation demonstrates an urgent requirement for **intelligent security systems** which use artificial intelligence to deliver ongoing security protection for educational institutions.

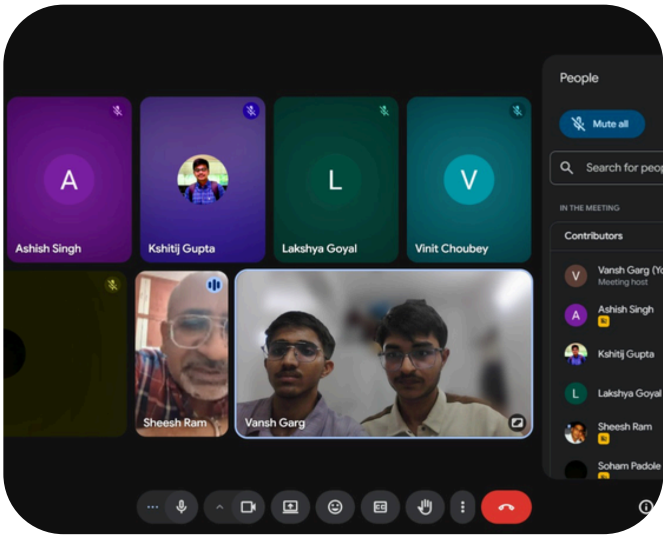
Interview with DRDO Scientists

For the purpose of evaluating and improving our AI-based surveillance system for hostel safety, we presented our project to two senior scientists from the **Defence Research and Development Organisation (DRDO), Jodhpur**. We shared our system's core features person detection, tracking, re-identification, and a risk-based alert system and sought their expert feedback on its real-world feasibility and design.
The interviewees were **Mr. Sanjay Sharma**, Scientist F, an expert in Surveillance and Security Systems, and **Mr. Sheesh Ram**, Scientist F, an expert in Antenna and Radar Detection Systems. Both bring decades of experience in defense-grade technology, and their insights played an important role in helping us refine the direction of our project.
Interviewees:

Mr. Sanjay Sharma — Scientist F, DRDO Jodhpur (Surveillance and Security Systems)

Mr. Sheesh Ram — Scientist F, DRDO Jodhpur (Antenna and Radar Detection Systems)





Highlights of the Interview

1. REAL-WORLD CHALLENGES IN AI DEPLOYMENT

Mr. Sanjay Sharma explained that while most current systems rely on simple motion detection to trigger recording, true AI-based identification is an evolving challenge. He thinks the transition from a lab setting to a busy hostel environment is difficult because:-

- **Dataset Integrity:** He said that training data is the most critical factor; without a diverse dataset, the system will **fail in real conditions.**
- **Angle of Approach:** He noted that unknown individuals can approach from any angle, requiring the system to be trained on multiple orientations, not just front-facing shots.
- **Active Avoidance:** He thinks that people with malicious intent will actively **try to hide their identity,** unlike authorized users who cooperate with the system.
- **Crowd Noise:** High-density conditions introduce "noise," making it significantly harder for an algorithm to maintain accuracy.

2. HARDWARE SETUP VS. SOFTWARE INTELLIGENCE

The scientists emphasized that even the best software cannot compensate for a **poor physical setup**, as system performance depends heavily on **input quality**. Mr. Sanjay Sharma suggested that cameras should be placed **directly in front of entry points**, slightly above the gate, rather than at high or angled positions to ensure **clear footage**. He also advised keeping the **system simple** in the initial stages, noting that AI models are **resource-intensive**. Additionally, he highlighted the importance of testing the system in **real-world**, **high-traffic environments** to ensure **reliability under practical conditions**.

3. THE MACHINE ADVANTAGE: FATIGUE AND EMOTION

A big part of the conversation centered on a simple but important question, can a machine guard better than a human? **Mr. Sanjay Sharma** thought so, and his reasoning made sense. People get tired, emotions cloud judgment, and anyone can be manipulated on a bad day. A machine simply doesn't have those vulnerabilities.

That said, he wasn't suggesting anyone flip a switch overnight. His idea was to let the AI work alongside human guards first, earn its place, and only take the wheel once it's hitting somewhere between **90 and 95 percent** effectiveness, a measured, confidence-building handover.

The scientists brought a grounded perspective to the table as well. Too many false alarms **don't just annoy people,** they make residents stop taking the system seriously. And once that trust is gone, even a perfect alert goes ignored. Getting the accuracy right wasn't optional; it was the whole point.

4. EXPERT SUGGESTIONS FOR SYSTEM IMPROVEMENT

Mr. Sanjay Sharma and Mr. Sheesh Ram suggested several approaches to improve the robustness of the system. They emphasized the importance of using **multi-feature identity representations** rather than relying solely on facial recognition, noting that characteristics such as **gait** can provide **additional cues for identification** under challenging conditions.

They also recommended incorporating a classification mechanism to distinguish between authorized and potentially suspicious individuals based on learned patterns. To improve reliability, they suggested a **hybrid detection approach**, combining **motion-based detection** with **identity recognition**, so that failure in one component can be compensated by the other.

Additionally, they proposed the idea of an **interconnected camera network**, where information about detected individuals can be shared across cameras to maintain **continuity** and improve responsiveness. Finally, they highlighted the need for user cooperation in surveillance systems, suggesting that **avoidance behavior**, such as consistently evading camera visibility, could be treated as a potential **indicator of suspicious activity**.

5. PRIVACY AND DATA SOVEREIGNTY

Addressing our concerns about ethics, the experts were quite reassuring. They said that as long as the data is stored locally within the hostel premises and **not put on the internet**, the system is secure and the risk of misuse is negligible. They think that once residents see the system is accurate and fair, any initial discomfort with surveillance will naturally fade.

6. SCALING AND MAINTENANCE

The experts emphasized that the development of such systems is inherently **iterative** and that no system achieves perfect performance at the outset. They noted that initial effectiveness may be limited, and improvements must be made gradually through **continuous** refinement. They highlighted the importance of **ongoing maintenance** and **regular data updates**, stating that system performance depends not only on deployment but also on **sustained improvement** over time. Additionally, Mr. Sanjay Sharma suggested the use of **covert** or less visible surveillance mechanisms to detect individuals attempting to **bypass primary monitoring systems**.

CONCLUSION

The scientists concluded that our project is very useful and could save significant time and manpower. Their final word of advice was to focus heavily on the testing phase.

There is always a scope of improvement... You have to invest most of your time in testing. That is my submission."

Mr Sanjay Sharma, DRDO

Stakeholder Interaction and Insights

STAKEHOLDER INTERACTION AND INSIGHTS

As part of understanding the real-world challenges associated with surveillance and security, we interacted with multiple stakeholders including security personnel, administrative authorities, infrastructure representatives, and technical mentors. These interactions helped us ground our system design in practical constraints rather than purely theoretical assumptions.

SECURITY GUARDS

Interactions with hostel security guards revealed that while the entry point of the hostel is actively monitored and access is generally controlled at the gate, their role is largely limited to entry-level supervision. Guards make decisions based on observation and judgment at the entrance, but there is **no mechanism to track individuals** once they move inside the premises. As a result, movement across floors, corridors, and rooms **remains unmonitored**. This leads to several practical challenges, including the inability to **monitor internal movement**, the absence of any system to detect repeated or **unusual patterns**, and the fact that suspicious behavior is often identified only **after an incident has already occurred**.

Additionally, the monitoring process is **entirely manual** and heavily dependent on human attention, which can vary due to workload and environmental conditions. While entry points are controlled, the absence of **continuous internal monitoring** and **automated alert mechanisms** creates a **significant security gap**. This highlights the need for a system capable of tracking individuals within the premises and providing real-time insights to assist security personnel in making timely decisions.

SECURITY OFFICER (AMIT RAWAT)

We interacted with the **institute's security officer**, who provided valuable operational insights. While he mentioned that he is not directly involved with AI-based systems, he highlighted several procedural and communication gaps in the current security workflow.
A key issue discussed was the **delay in official communication.** For instance:

It was observed that new faculty or staff may join earlier in the week, while official circulars confirming their identity are often released later. This delay creates **ambiguity for security guards**, making it difficult to reliably **verify individuals claiming to be faculty**. Additionally, insights from past incidents highlighted how factors such as **timing gaps**, **impersonation**, and the **absence of clear verification protocols** contribute to security vulnerabilities.

INFRASTRUCTURE PERSPECTIVE (PUNJAB STATE POWER CORPORATION LIMITED – PSPCL)

We also interacted with **Chief Auditor, PSPCL, CA Rajan Gupta**, to understand the infrastructural and operational considerations involved in deploying large-scale surveillance systems. The discussion highlighted that any continuous monitoring solution must be designed with a strong focus on **reliability** and **practical feasibility**. In real-world environments, system performance is highly dependent on factors such as **uninterrupted power supply** for cameras and processing units, as well as stable network connectivity to support real-time data transmission.

Additionally, **scalability** was identified as a critical requirement, particularly when extending the system across multiple buildings or larger institutional spaces. It was further emphasized that institutions must consider **cost, maintenance, and long-term sustainability**, rather than focusing solely on technical capability. These insights indicate that while AI-based surveillance systems can significantly enhance monitoring, their real-world effectiveness depends equally on infrastructure readiness and careful deployment planning.

SECRETARY AND MENTORS (IOTA CLUSTER - AI CLUB, IIT ROPAR)

We also interacted with mentors and the secretary of the **Iota Cluster - The AI Club,** student club at IIT Ropar, to gain technical perspective on building a practical and reliable system.

The discussions emphasized that while advanced models and techniques are important, real-world systems must prioritize **stability**, **robustness**, and **consistency** over complexity. It was noted that computer vision systems commonly face challenges such as variations in **lighting conditions**, **occlusion** between individuals, **similar appearances** among different people, and **inconsistent detections** across frames. To address these issues, it was suggested that systems should not rely on a single model, but instead **combine multiple components** such as detection, tracking, and identity matching to improve overall reliability.

Additionally, it was highlighted that real-world deployments require careful **parameter tuning** and **system-level optimization**, as performance can vary significantly depending on environmental conditions and camera configurations. These insights reinforced our approach of designing a modular pipeline that integrates detection, tracking, and re-identification, while maintaining stability and adaptability in practical scenarios.

KEY TAKEAWAYS

From the stakeholder interactions, several consistent themes emerged. Current surveillance systems are largely **reactive** rather than **proactive**, limiting their ability to prevent incidents in real time. It was also evident that **human monitoring** alone is insufficient **at scale**, especially in environments with multiple camera feeds. There is a clear need for **real-time alerts** and **decision support systems** to assist security personnel in responding more effectively. Additionally, practical deployment constraints, such as infrastructure and usability, were identified as being equally important as technical capability.

These insights directly influenced the design of our AI-based surveillance system, which focuses on **automated detection**, **tracking**, and **alert generation** while remaining grounded in real-world feasibility.

Field Study and Spatial Analysis

Effective surveillance in dense residential environments depends not only on the presence of cameras but on how well their **positioning** aligns with the spatial dynamics of the infrastructure. To evaluate the **effectiveness** of existing surveillance infrastructure and understand its spatial limitations, we conducted a detailed field study of the Chenab Hostel. This study focused on **analyzing** relationship between physical space, **camera placement**, and visibility coverage, with the objective of identifying gaps in monitoring and **potential vulnerabilities** in the system.

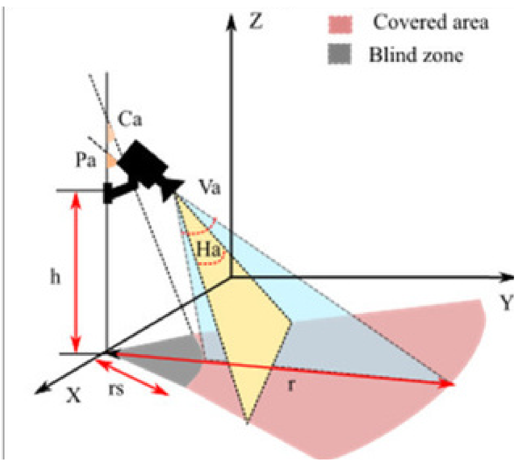
3D INFRASTRUCTURE MODELING

As a foundational step in **spatial reconstruction**, we developed a 3D model of the Chenab Hostel using **CAD based tools**.
The model included all the structural elements such as floors, corridors, staircases, rooms, and entry-exit points, with a high degree of geometric accuracy.
This model helped us to translate the real world environment into a digital space where spatial relationships could be studied systematically. Unlike conventional **2D layouts**, the 3D model included vertical constraints, angular orientations, and obstruction factors, parameters that significantly influence the surveillance performance but are often overlooked in planar analysis.

FIELD-OF-VIEW BASED COVERAGE ANALYSIS

To assess the monitoring capability of each camera, a field-of-view **(FOV) based analysis** was conducted in the 3D model. Each camera was modeled as a conical projection, having its own viewing angle and effective range.

The ground coverage of a camera was estimated using the **geometric relations** derived from its **height** and angular spread. For a camera placed at height 'h' with **angular limits (θ1 and θ2)** determined by its tilt and field-of-view, the visible region on the ground can be approximated by using trigonometric relations:



Let H_c be the camera height and ϕ be the tilt angle. Let θ_h and θ_v denote the horizontal and vertical field of view angles, respectively.

Distances to floor intersection:

$$d_{\text{near}} = H_c \tan\left(\phi - \frac{\theta_v}{2}\right), \quad d_{\text{far}} = H_c \tan\left(\phi + \frac{\theta_v}{2}\right) \tag{1}$$

Note: $d_{\text{near}} \geq 0$ (the camera cannot see behind itself).

Width of coverage at distance d:

$$\text{Width}(d) = 2d \tan\left(\frac{\theta_h}{2}\right) \tag{2}$$

Total floor coverage area:

$$A_{\text{room}} \approx \int_{d_{\text{near}}}^{d_{\text{far}}} 2d \tan\left(\frac{\theta_h}{2}\right) dd \tag{3}$$

After integration:

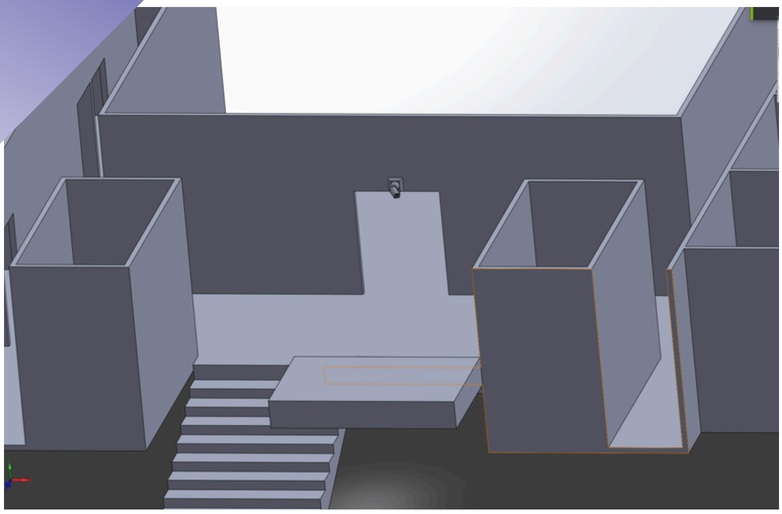
$$A_{\text{room}} = \tan\left(\frac{\theta_h}{2}\right)\left(d_{\text{far}}^2 - d_{\text{near}}^2\right) \tag{4}$$

Final compact expression:

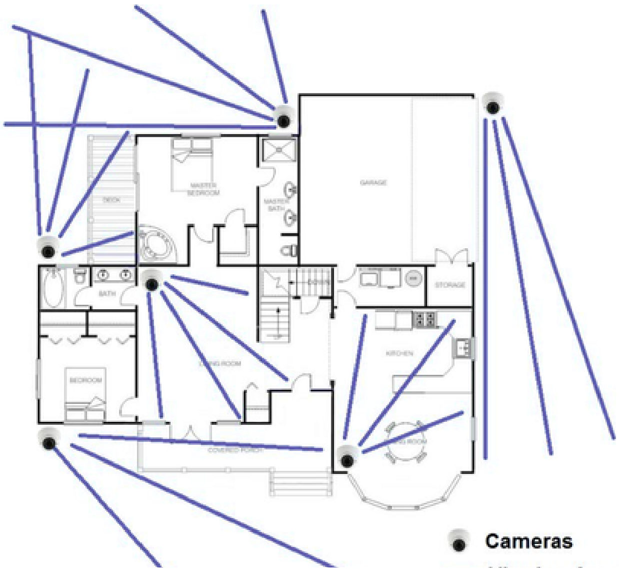
$$A_{\text{room}} = \tan\left(\frac{\theta_h}{2}\right)\left[\left(H_c \tan\left(\phi + \frac{\theta_v}{2}\right)\right)^2 - \left(H_c \tan\left(\phi - \frac{\theta_v}{2}\right)\right)^2\right] \tag{5}$$

This helped us to project an effective coverage region onto the ground. By extending this for all cameras in the **3D model,** we obtained an **overall visualization** of the monitored and **unmonitored** zones. This approach confirmed that the analysis was not assumption based but grounded in the **actual geometric constraints** of the system.

Detection of Blind Spots and Choke Points



Mapping camera coverage directly onto the **spatial model** revealed several unmonitored gaps i.e **'Blind Spots'**. These blind spots are primarily concentrated in **transition areas** such as staircases, entrances and corridor junctions.



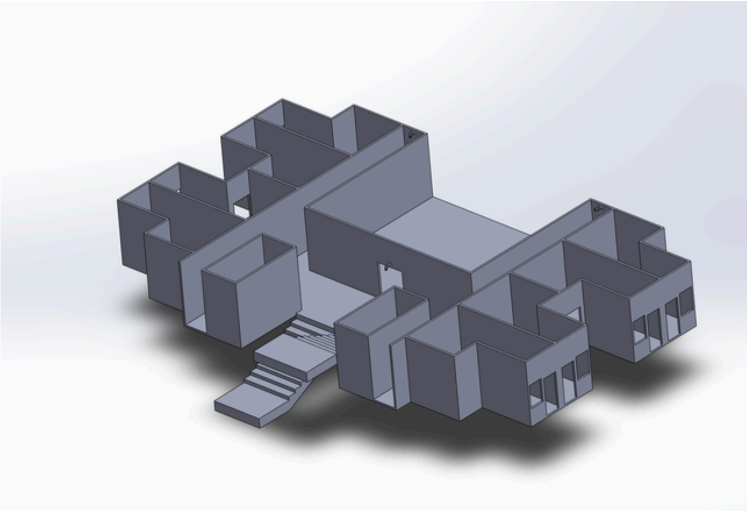
The study also revealed 'choke points', these are **narrow passages** and main entryways where foot traffic naturally clusters and **density increases.** More concerning, however, is that many of these **high risk zones** lack consistent visibility, meaning they are either partially or entirely **unmonitored**.

This lack of coverage **breaks** continuous surveillance. When individuals move through these **unmapped regions** without being tracked, it creates a significant **security gap,** making it very hard to maintain **real time monitoring.**

Key Insights and Implications

The field study demonstrates that surveillance **effectiveness** is basically a spatial problem rather than a pure technological one. Accurate monitoring requires an **understanding** of how individuals move through space, where bottleneck forms, and how visibility is affected by **structural design.**
The integration of CAD based modeling with geometrical coverage analysis gave a precise and **scalable method** to **evaluate** surveillance systems.

More importantly, it allowed identification of infrastructure gaps that would otherwise remain undetected through conventional observation.



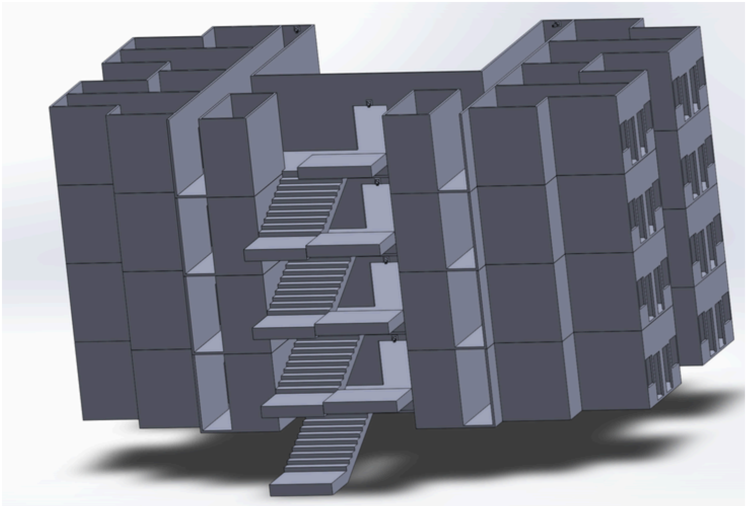
Conclusion of Analysis

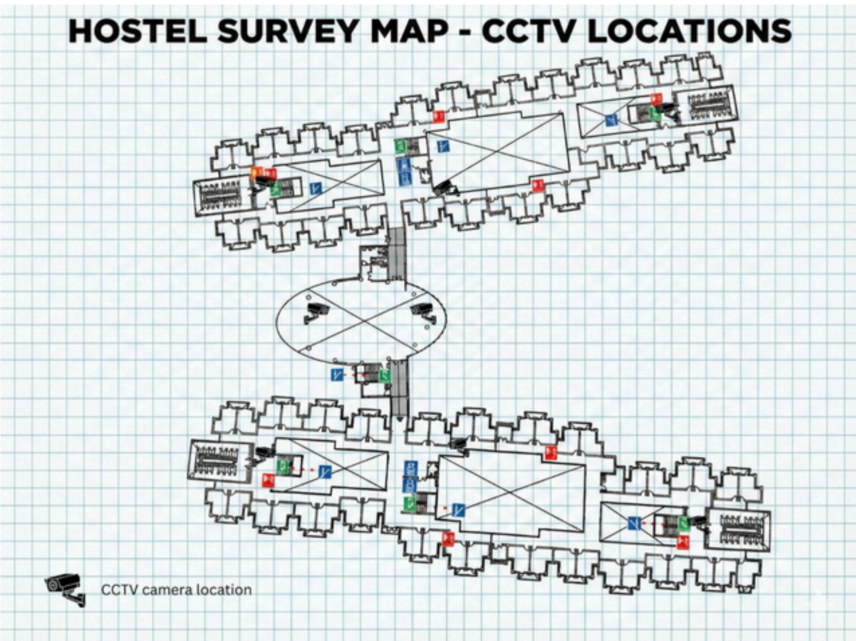
Through the combined use of 3D modeling, camera mapping, and trigonometric coverage estimation, this study provided a **clear understanding** of surveillance **distribution**, with an example of the Chenab Hostel. The identification of blind spots and choke points demonstrates present **weaknesses in the current system** and emphasizes the need for a more structured, **spatially aware approach** to camera **placement.**

CAMERA MAPPING AND PLANAR VISUALIZATION

After the **spatial reconstruction**, the positions of all the installed surveillance cameras within the hostel were mapped during the study.

This was achieved by **CAD** based modelling tools, a modelling software used in many high end studies for **systematic** spatial analysis. These positions were then plotted onto a 2D top view layout, providing a **clear visual** representation of camera distribution across different floors and areas.





This mapping showed that while cameras were placed at most of the **key locations** such as corridors and entrances, their placement appeared to be functionally **independent** instead of being part of a **coordinated** spatial network. The absence of a **unified placement** strategy indicated possible discontinuities in surveillance coverage.

DETAILED DESCRIPTION OF THE PROBLEM

In any large educational institution, security is usually handled by a network of **CCTV cameras.** However, through our research and observations at the hostel facilities, we found that simply **having cameras** does not mean the area is secure. There are several deep-rooted problems in how surveillance currently works.

The Passive Nature of Current Systems

One of the biggest issues is that most **CCTV** systems are **passive** . They only record footage and store it, but they don't actively analyze what is happening.
A real example of this can be seen from an incident at **Osmania University** (2024), where an outsider entered a girls' hostel at night and tried to steal a phone. Even after threatening a student, he managed to escape. The cameras **recorded everything**, but they didn't help during the incident.
This shows that current systems mostly help after something goes wrong, not before. So **instead of preventing** crime, they are just used to check what happened later.

LACK OF CONNECTION BETWEEN CAMERAS

Another major issue is that cameras work **independently.** Each camera acts like a separate unit and doesn't share information with others.
In a hostel with multiple floors and corridors, a person keeps moving from one **camera view** to another. But the system cannot track that it is the same person.
Because of this:

- There is **no continuous** tracking of a person
- Suspicious people can **easily move** unnoticed
- The system cannot differentiate between residents and outsiders

So overall, the footage becomes **disconnected** pieces instead of a **continuous flow**.

DEPENDENCE ON HUMAN MONITORING

Right now, **security guards** have to monitor all the screens. But honestly, this is not practical.
Watching many screens continuously for long hours is **tiring.** After some time, attention naturally **drops**. This is called **monitoring fatigue**.
It is scientifically impossible for a human to watch **50** screens for an 8-hour shift. After about **20** minutes, the brain gets tired and starts missing up to 90% of the movement on the monitors.

BLIND SPOTS IN INFRASTRUCTURE

Even with **many cameras**, there are always some areas which are not covered properly. These are **blind spots**
For example:

- Staircases
- Corners
- Back exits

Also, cameras are often placed at **higher positions** to cover more area. But because of that, faces are not clearly **visible**.
So even if someone is seen once, once they move out of that view, it becomes **difficult to track** them properly.

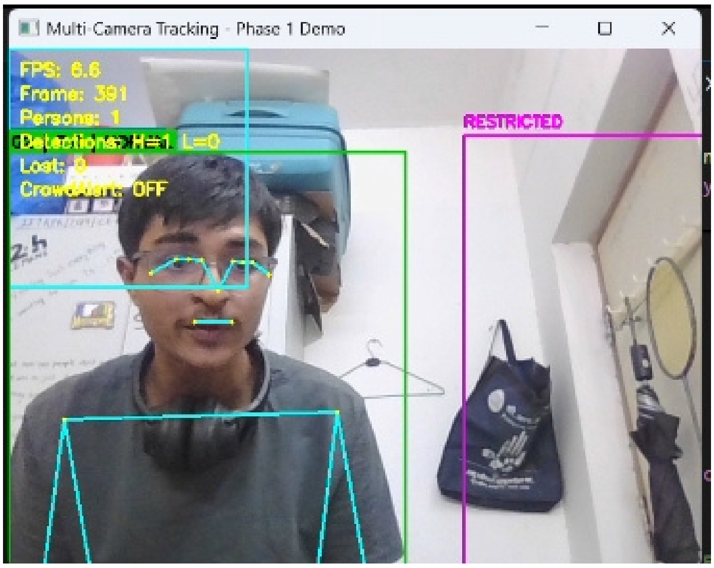
HOW ARE WE DIFFERENT ?

Standard Industry Tech: Most places use basic **IP cameras** that have **"motion detection"**. But in a hostel, there is always motion—students walking to get water or going to the washroom. This causes too many **false alarms**, so guards eventually turn the volume down or ignore them.

Global AI Trends: Advanced cities like Shanghai and Singapore use **Computer Vision** to track movement across entire blocks. They use models like **YOLO** to count people and spot unusual behavior in real-time.

Our Innovation - Camera Geometry: One thing we are doing differently is using **math and trigonometry** to make cameras smarter. Instead of just guessing, our system calculates the exact ground area a camera sees based on its height and angle. This lets us set **"Dynamic Thresholds"** so the system knows exactly when an area is too crowded or if someone is where they shouldn't be

Real-Time Dashboard: Unlike old systems where you just watch video, we are building a **web portal.** It acts as an all-in-one dashboard that shows a "heat map" of where people are and sends **instant alerts** if a threat is detected



NEED AND SIGNIFICANCE OF THE PROJECT

Resolving the gaps in our surveillance is a social necessity that affects the daily lives of the entire student community.

1 STUDENT SAFETY AND COMFORT

A hostel is not just a building, it is where students live and keep their important belongings. When **theft** happens, it creates **fear** and **discomfort**.
Students start becoming extra **cautious**, and the overall environment changes. So there is a need for a better system that can make students feel **safe** again.

---

2 MOVING FROM REACTIVE TO PREVENTIVE SYSTEM

Current systems mostly react after something **happens**. But ideally, we **should prevent** incidents before they occur.
With **smarter** systems like AI-based **tracking** and **alerts**, it is possible to detect **suspicious** activity early and take action in time.

---

3 BETTER USE OF RESOURCES

Security guards already have a lot of **responsibility**. They cannot monitor everything all the time.
With **automation**:

- Their workload reduces
- Response time **improves**
- **Monitoring** becomes more efficient

Also, improving existing cameras using **software** is more cost-effective than installing completely new systems.

FUTURE SCOPE

4

This idea is not **limited** to hostels only. It can be **expanded** to other areas like:

- Campus security
- Parking areas
- Public spaces

It can be a step towards building a **smart campus** where **technology** and **human effort** work together.

5

PREVENTING PANIC SITUATIONS:

Sometimes even a small **rumor** or **misunderstanding** can create panic in crowded places. We have seen such situations where **confusion** leads to **bigger** problems. In a hostel also, a **false** alarm about an **intruder** can cause chaos. Our system helps by giving accurate and real-time information so that such **panic situations** can be avoided.

IMPROVING COLLEGE REPUTATION:

6

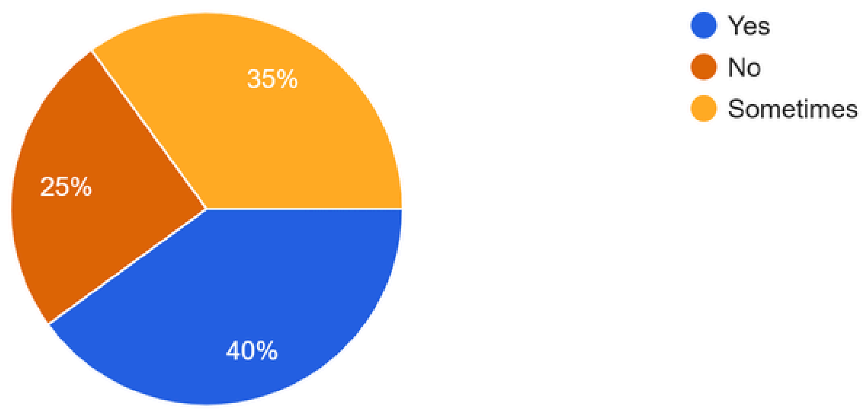
For an institute like **IIT** Ropar, safety is also important for maintaining its **reputation**. A secure campus attracts good students, faculty, and even **international** collaborations. With our project (GuardianPath), we are trying to show that students can use their **technical** knowledge to solve **real-life problems** within the campus.

Survey Analysis

We conducted two surveys to understand perceptions of **safety** and the effectiveness of **existing surveillance systems**. One survey targeted individuals within the campus **(260 responses)**, while the other collected responses from a broader audience outside the campus **(1670 responses)**. This allowed us to analyze both localized and general perspectives on security and surveillance.

SAFETY PERCEPTION

Have you ever felt unsafe even when a CCTV camera was present?
260 responses

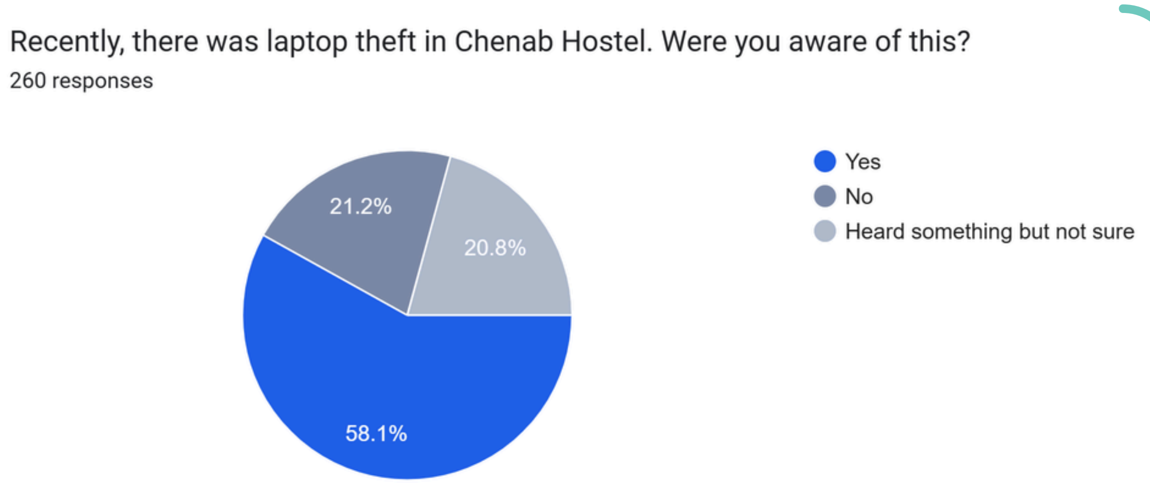


Survey Response to "Have you ever felt unsafe even when a CCTV camera was present?"

A significant proportion of respondents reported feeling unsafe even in the presence of CCTV cameras. Approximately **40% felt unsafe**, while **35% reported feeling unsafe at times**,indicating that the presence of cameras alone does not guarantee a sense of security.

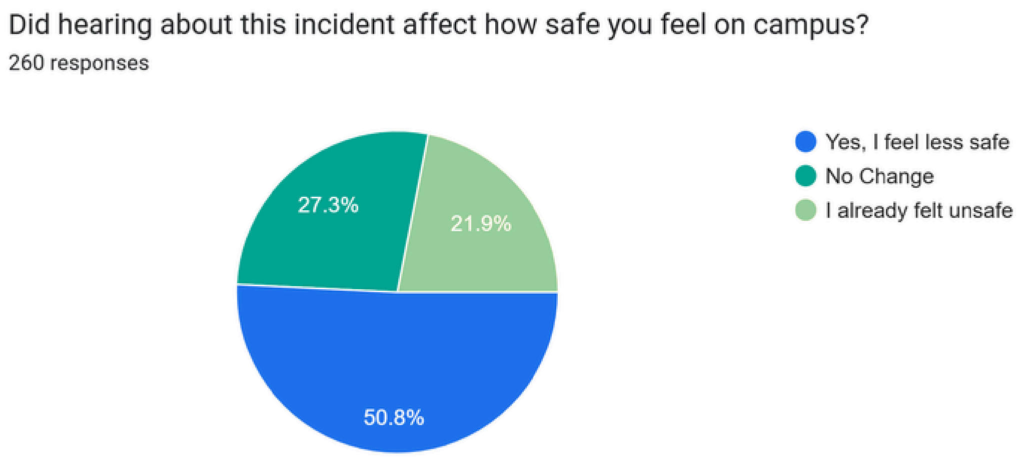
Similar trends were observed in the broader survey **(1670 responses)**, reinforcing that this issue is more widespread.

INCIDENT ANALYSIS



Survey Response to "Recently, there was laptop theft in Chenab Hostel. Were you aware of this?"

A majority of respondents **(58.1%)** were aware of the recent **laptop theft incident** in Chenab Hostel, indicating that such events have a strong presence in **public awareness.**

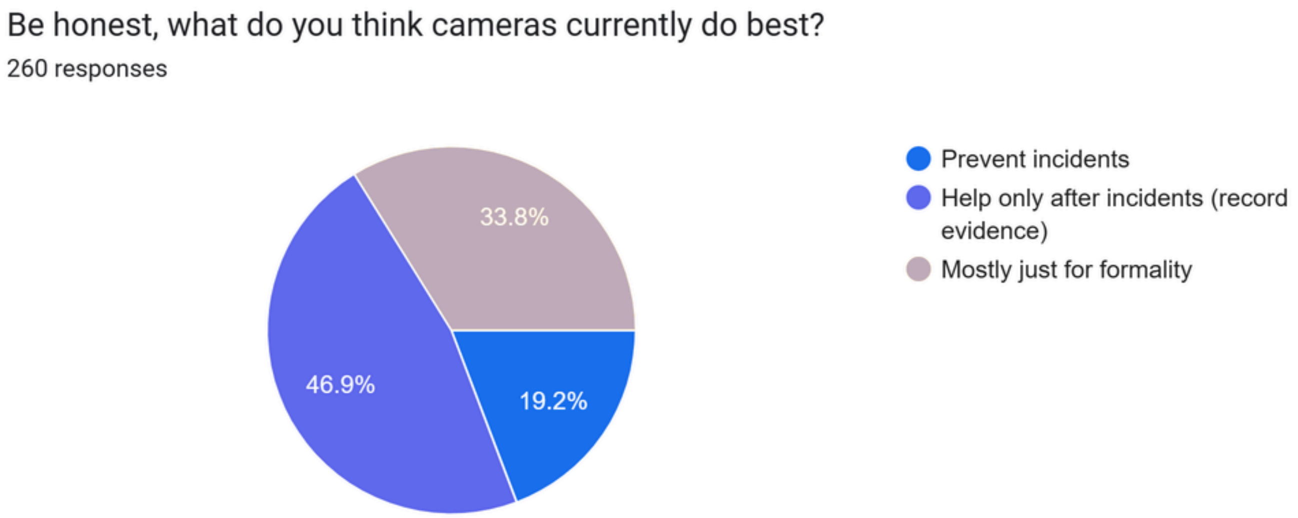


Survey Response to "Did hearing about this incident affect how safe you feel on campus?"

More than half of the respondents **(50.8%)** reported feeling **less safe** after hearing about the incident, while (**21.9%)** stated that they **already felt unsafe**. This highlights the significant impact real incidents have on perceived security.

These findings show that **real-world incidents** play a **crucial role** in shaping how safe individuals feel on campus.

CCTV EFFECTIVENESS



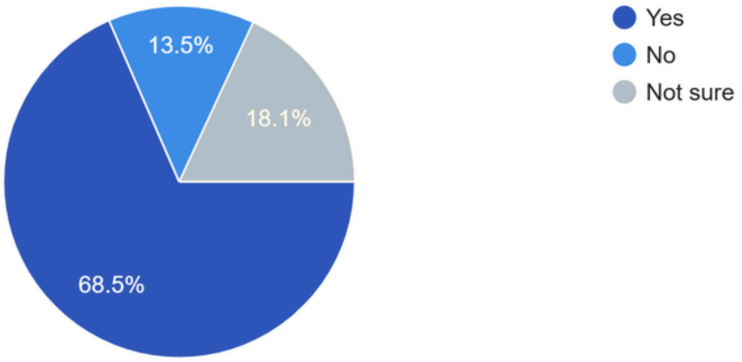
Survey Response to "Be honest, what do you think cameras currently do best?"

A large proportion of respondents **(46.9%)** believe that CCTV systems are primarily useful **after incidents occur**, mainly for **recording evidence**. Additionally, (**33.8%)** feel that cameras are **mostly for formality**, while only (**19.2%)** believe they **actively prevent incidents**.

This clearly indicates that current CCTV systems are perceived as **reactive rather than preventive**, highlighting a **major limitation** in existing surveillance infrastructure.

NEED FOR OUR SYSTEM

Would you feel safer if a system could detect suspicious or inappropriate activity in real time and alert security immediately?
260 responses



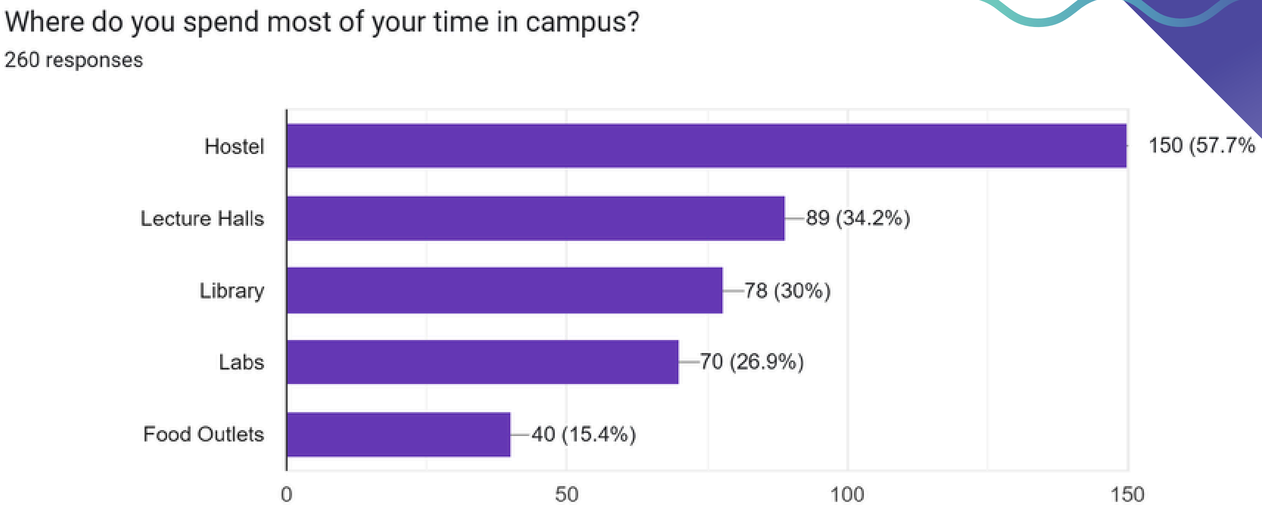
Survey Response to "Would you feel safer if a system could detect suspicious or inappropriate activity in real time and alert security immediately"

A strong majority of respondents **(68.5%)** indicated that they would feel safer if a system could detect suspicious activity in **real time** and **alert security immediately**. This reflects a clear preference for **proactive safety measures** rather than passive monitoring. It also shows that individuals value **faster response** and early intervention in potentially risky situations.

Similar trends were observed in the broader survey **(65.3% agreement)**, reinforcing the demand for **proactive** surveillance systems. This consistency across different respondent groups suggests that the need for such systems is not limited to a single environment but is **widely recognized**.

This demonstrates a **clear need** for **intelligent**, real-time monitoring systems that go **beyond passive recording**. Such systems can assist security personnel by enabling quicker **decision-making** and improving overall **response efficiency.**

SPATIAL ANALYSIS & FINAL INSIGHTS



Survey Response to "Where do you spend most of your time in campus"

The majority of respondents **(57.7%)** reported spending most of their time in **hostels**, followed by lecture halls (34.2%) and libraries (30%). This indicates that hostels are critical zones requiring enhanced monitoring due to **high occupancy**.

The survey highlights several important insights. A significant number of individuals feel unsafe despite the presence of CCTV systems, indicating a lack of trust in existing infrastructure. Surveillance systems are widely perceived as **passive** and **reactive**, rather than proactive. **Real-world incidents** strongly influence perceptions of safety, often exposing system limitations. Additionally, high-occupancy areas such as **hostels** require more focused monitoring. Overall, there is a clear demand for systems capable of **real-time detection and alert generation**.

Survey datasets used for analysis are provided below for reference:

Campus Survey Dataset **(260 responses)**:
https://docs.google.com/spreadsheets/d/1cORfkFmjIl4cy7_i_Fk1tsmePrkYw9vMdJdU6tv3qKA/edit?usp=sharing

External Survey Dataset **(1670 responses)**:
https://docs.google.com/spreadsheets/d/1JDDhPbfa1fBG-gSQpzeIdu9A8k-8X3UJ7nThocPHIB8/edit?usp=sharing

OBJECTIVE

In order to solve the issue of **security vulnerabilities** revealed by the theft incident in the hostel and also the **general inefficiency** of passive surveillance systems, it was imperative for us to introduce a technology-driven solution. The **objectives** of the proposed system are listed below:

1

To Ensure Continual and Continuous Person Tracking

The first main **goal** of the system consists of assigning each person observed by the camera with a **constant identity** and **tracking** it throughout the observation session , through occlusions, crowdedness, short disappearance and reappearance. **Contrary** to traditional passive surveillance technologies that only record data, our system ensures continual and **uninterrupted tracking** of each and **every person** including their behaviour and movement.

2

To Ensure Cross-Camera Re-Identification

A single-camera system is **not** enough for use in a **multi-building campus**, because our **system** should understand that the person identified as identity #4 on Camera 1 is the **same as** identity #16 on Camera 3, basing solely on the image of the target. It is cross-camera **identity continuity** that allows treating **multiple cameras** as an **integrated system** and work as a dynamic cointegrated team.

3

To Generate Immediate Alerts and Event Responses

Security issues occur spontaneously and require an **immediate response**. The proposed system should provide **timely** generation of real-time alerts upon each **significant event** such as entrance into prohibited areas, identification of a previously disappeared identity that was **suspicious** or **unusual** movements – since every second counts in such cases.

4

To Offer Centralized Real-Time Monitoring System

The web-based operator **console** of the surveillance system would gather information about each person, his **movement routes** and events in one single interface. It will allow security personnel to **manage** the entire campus through a **single window** with real-time data and maps that include the movement track of **all people**.

Tools & Techniques

This project attempts to solve one operational issue in particular, that is, who is in each frame across a network of cameras, where are they, and is the person currently visible on Camera 3 the same person who was seen on Camera 1 earlier? To solve this problem, there are needed four different technical tools that solve four specific sub-questions.

Person Detection

The very first part in the pipeline is the **detection** of every single person in every single frame. We use one of the best models that exists for such purposes, **YOLOv11m**, that operates on an input image resolution of **1280 px.** Why exactly this resolution is chosen? Because persons located far away, partially **occluded,** or in low-confidence areas in the frame are detectable only at such resolution; otherwise, they would **not** be detected **at all.** Moreover, our model outputs only detections of persons; also, **two confidence** mechanisms are utilized here.

Within-Camera Tracking

After people have been successfully detected in a frame, we need to **assign** an identity to each individual and **track** this identity across multiple frames in the future. Such task is accomplished through the utilization of two different tools at once which are **Kalman Filter** and the **Hungarian Algorithm**. In short, we use the Kalman Filter to predict future **positions** of every tracked person according to their last location, speed, and change in size; then, we apply the Hungarian Algorithm to **match** new detections to these **predicted** points in space. ..

Cross-Camera Re-Identification

Intra-camera person tracking will not suffice in identifying whether the same individual moved to another camera area or not. We utilize **OSNet_x1_0** that is trained on **MSMT17** – the largest and most heterogeneous multi-camera person re-identification dataset. Each confirmed person's visual signature will be well represented by a **vector of 512** dimensions that **represents** the person's clothing, colours, texture, and body shape which is then **compared** against an entire **set of identities** seen across all cameras and a successful matching over the similarity **threshold** will result in a cross-camera person **re-identification.**

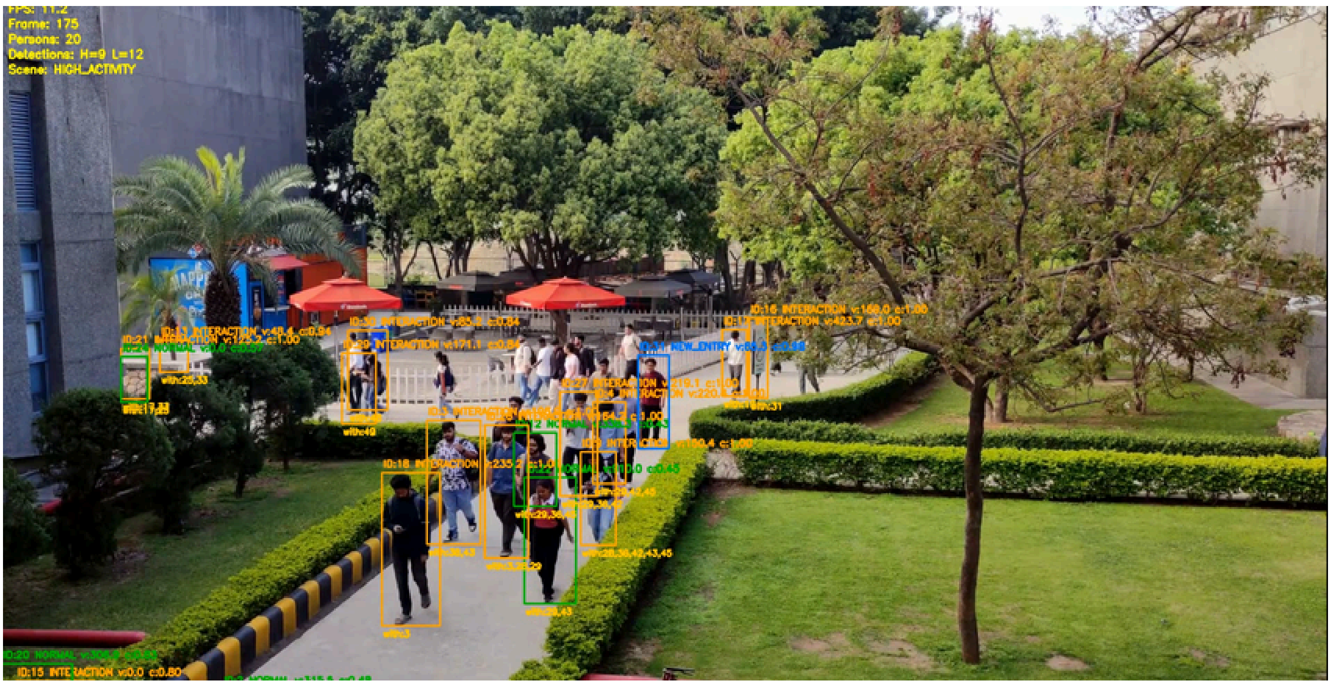
Dashboard and Alert System

Tracking, re-identifications, and videos coming from cameras are **streamed** via Socket.IO WebSocket to a web application that is built using **React framework** in real-time. This web app allows the user to **view** camera videos and their **annotations,** see the geographical movements of people through camera areas using the geographic **map**, view **timestamped events** and get analysis of the system in a **unified interface**. An occurrence of an important event can **trigger** the alert system that will alert operators with the **details** of the event immediately.

Proposed Intelligent Multi-Camera Surveillance System

Traditional surveillance systems operate in a **passive mode** which records events without detecting any suspicious activities. Our proposed solution introduces an AI-powered intelligent surveillance system which can **detect** and **track** people through **multiple cameras** while it maintains **real-time alerts** for security personnel.

- Intelligent replacement for passive CCTV systems
- Real-time person detection and tracking
- Cross-camera identity matching
- Suspicious activity detection
- Instant alert generation
- Centralized operator dashboard



ENTRY MONITORING AND TRACKING

The system uses the **YOLOv11m model** at 1280p resolution to accurately detect people entering surveillance zones, including distant or partially occluded individuals. The system uses identity assignment to track each detected person through the combination of **Kalman Filters** and the **Hungarian Algorithm.**

- YOLOv11m used for person detection
- High-resolution detection improves accuracy
- Handles occlusions and crowded scenes
- Assigns **unique IDs** to individuals
- Uses Kalman Filter for **motion prediction**
- Hungarian Algorithm used for track matching

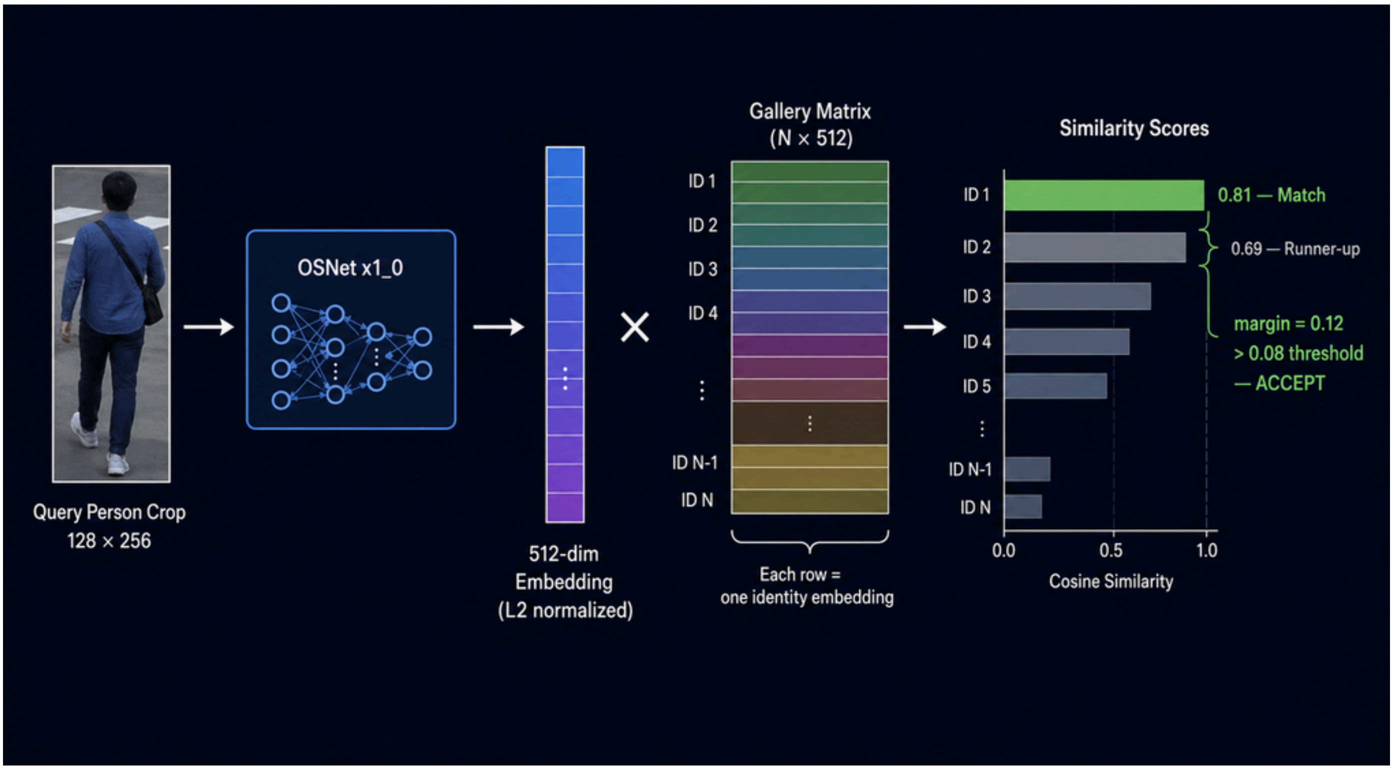
DETECTION STRATEGY

- High confidence detections (**conf ≥ 0.40**)
- Low confidence detections (**0.10 ≤ conf < 0.40**)
- Prevents false negatives during occlusion
- Avoids ghost track generation2

High confidence detections (**conf ≥ 0.40**) create reliable detection results. The system detects low confidence level detections when the confidence value falls between **0.10** and **0.40**. The system prevents false negatives that occur during occlusion events. The system prevents the creation of ghost tracks.

IDENTITY MATCHING AND MOVEMENT MAPPING

The system maintains identity continuity across multiple cameras using **OSNet_x1_0**-based person re-identification. The system converts each detected person into a feature embedding which it uses to match them against a **global identity database.** The system archives all movement events to create complete records of movement trajectories throughout the campus.



- **Cross-camera** identity matching
- Uses **OSNet_x1_0** deep learning model
- Generates **512-dimensional** feature embeddings
- Uses **cosine** similarity for matching
- Maintains global identity continuity
- Stores movement **timelines and paths**
- Displays trajectories on campus maps

SUSPICIOUS ACTIVITY DETECTION AND ALERTS

The system uses automated analysis to examine tracking data and detect any activities that **appear suspicious** which it uses to create alerts that are sent out in real time. Security personnel receive alerts through an **event-driven pipeline** which delivers them instantly.

- Detects **abnormal** movement behavior
- **Lost-and-unresolved** identity alerts
- Loitering detection
- Priority alerts for suspicious individuals
- Real-time event processing
- Sub-second alert delivery using **Socket.IO**
- Alerts include **timestamps, camera IDs, and thumbnails**
- Notifications visible only to security personnel

OPERATOR DASHBOARD AND CONCLUSION

A security operator dashboard built with React framework displays **real-time monitoring** capabilities and tracking of events and movement detection and analytics functions. The proposed solution transforms conventional **CCTV systems** into an intelligent and proactive surveillance platform.

DASHBOARD FEATURES

- **Live Camera Grid** with tracking overlays
- Tracking Map showing **movement paths**
- Event Log with **alerts and timestamps**
- Analytics for tracking and detection statistics

CONCLUSION

The proposed intelligent surveillance system transforms traditional **CCTV monitoring** into a proactive and **AI-driven security solution.** The system enables security personnel to monitor suspicious activities through its combination of **real-time person detection** multi-camera tracking cross-camera **re-identification and automated alert generation** capabilities. The solution achieves maximum effectiveness for campus and **large-area surveillance** environments through its movement mapping and centralized analytics system.

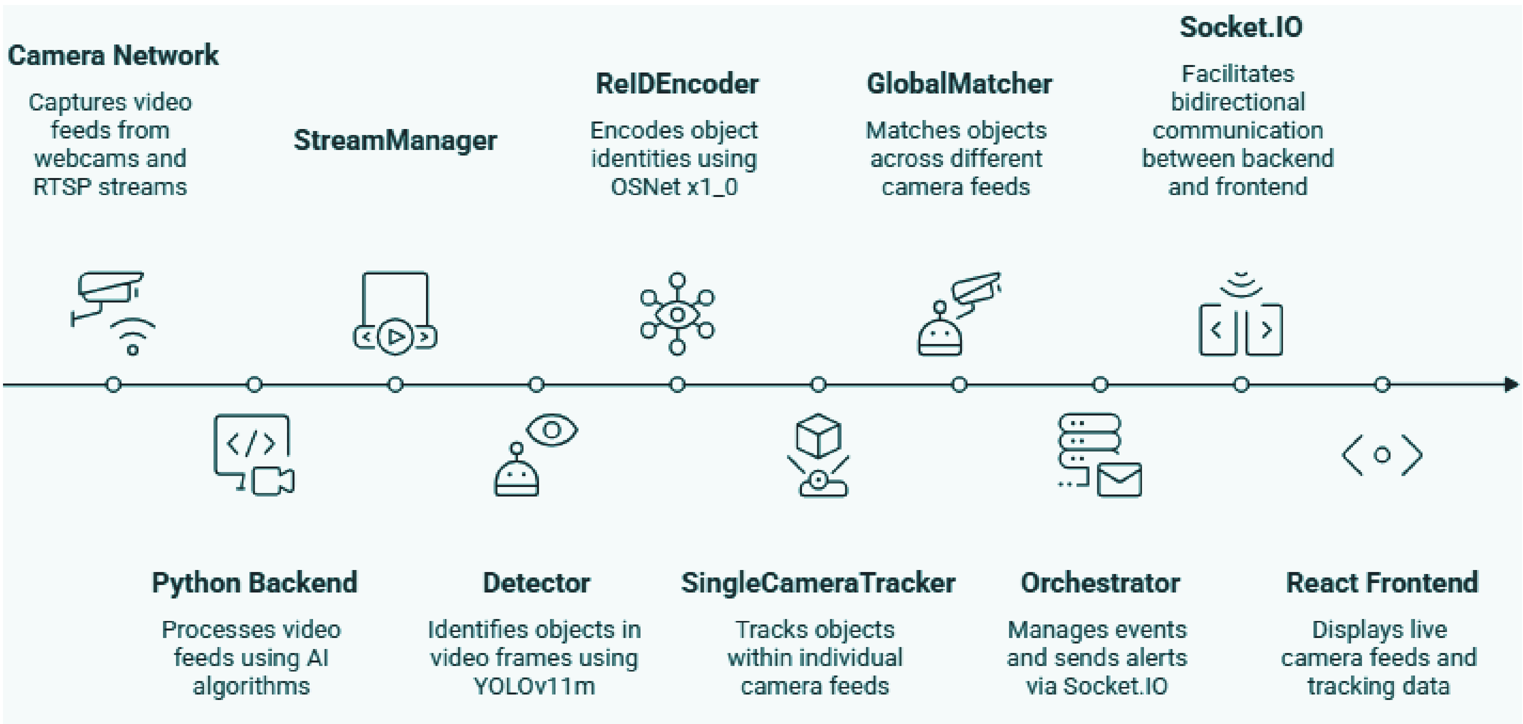
- **Intelligent real-time surveillance system**
  The system performs ongoing analysis of **live camera feeds** while it automatically identifies key events through its detection system which operates **independently** from manual observation.
- **Accurate multi-camera tracking**
  Individuals are reidentified as they are tracked from **one camera to another,** so that they remain identifiable throughout this network of surveillance.
- **Faster incident response and investigation**
  Providing revisions or discrepancies for **real-time alerts** and reconstruction of trajectories allows prompt responsiveness on the part of **security staff members, helping effective incident investigations.**
- **Improved campus security and monitoring**
  The system improves campus safety because it detects **suspicious activities** through its ability to monitor loitering and unauthorized movement and blind-spot disappearances.
- **Scalable and efficient solution**
  The design of the architecture would support such improvements as **multiple cameras** and **more prominent surveillance environments** while maintaining accurate non-lagged timings.

System Architecture

The system is built in two tiers , a Python backend that does all the AI heavy lifting, and a React frontend that security personnel actually interact with. They talk to each other over a persistent **Socket.IO WebSocket connection**, which means events flow from the backend to the dashboard in under a second. No polling, no refresh buttons.

What makes the architecture work is that every component has one job and does only that job. The stream manager handles cameras. The detector finds people. The encoder describes them. The tracker follows them. The global matcher connects them across cameras. When each module stays in its lane, the whole pipeline stays debuggable , a failure in re-identification **doesn't corrupt the tracker**, and a dropped camera doesn't crash the system.

The diagram below shows how data moves through the system from camera input to operator dashboard:



Overview : How Data Flows Through the System

Every processing cycle starts with the StreamManager pulling the latest frame from each camera. These frames go to the detector as a single batched GPU call , not one camera at a time, all of them together. The detector returns **bounding boxes** and **confidence scores**. High-confidence detections go to **OSNet** for appearance encoding. Low-confidence ones are kept aside as motion anchors.

Each camera has its own **SingleCameraTracker** instance. The tracker predicts where existing tracks should be, runs the matching cascade, updates confirmed tracks with **new measurements**, and **emits state changes** as events. Once per cycle, the GlobalMatcher collects the active tracks from all cameras and checks whether anyone on Camera 2 matches a known identity from Camera 1.

Confirmed events , appearances, re-identifications, losses , go to the Orchestrator, which routes them to the frontend over **Socket.IO** and **triggers alerts** where needed. The frontend updates live. The whole cycle runs continuously, camera to dashboard, with no human intervention required.

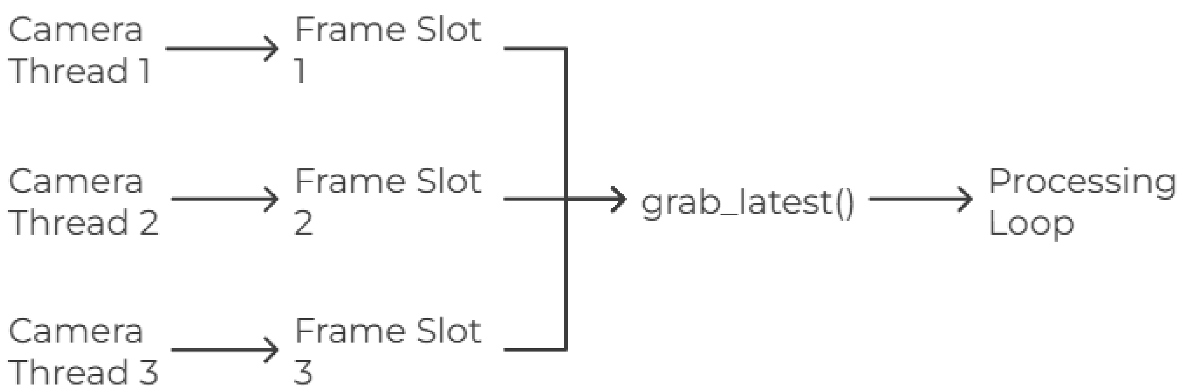
The key throughput decision is batched GPU inference. With four cameras running, naively processing them sequentially would mean eight GPU calls per cycle , four for detection, four for encoding. Batching collapses that to two. At four cameras, this alone saves roughly 1**8ms per cycle from kernel-launch overhead,** independent of what the models actually compute. At scale, it matters.

Camera Layer , Stream Manager

The **camera layer** is simpler than it sounds, but it has one **design choice** that everything else depends on: **the single-slot frame buffer**.

Each camera runs in its own **daemon thread**. The thread does nothing but **capture frames** as fast as the source allows and write the latest one into a shared **memory slot**. When the main processing loop calls `` `grab_latest()` ``, it gets whatever is in that slot right now , not the frame from two cycles ago, not a queued backlog. The most recent one.

Camera Frame Acquisition and Processing



This means the system never falls behind on **live streams**. If a particularly **complex frame** takes 300ms to process, the next call skips everything that arrived in between and picks up from now. For security monitoring, a slightly stale frame is far better than a growing backlog.

Video files work differently, and deliberately so. The **capture thread** waits for the main loop to signal that it has finished processing before advancing to the next frame. This ensures every frame gets **processed in order** , nothing skipped. For forensic replay of recorded footage, completeness matters more than real-time speed.

RTSP network cameras get **automatic reconnection**. If the network drops, the thread retries every two seconds and returns `None` in the meantime. The main loop sees a missing camera, skips it for that cycle, and keeps processing everything else. The system degrades gracefully , one bad camera doesn't take down the rest.

All **camera threads** are **daemon threads**, which means Python cleans them up automatically when the main process exits. No orphaned processes, no manual cleanup required.

Backend AI System , Detector and ReID Encoder

The Detector

Detection runs on **YOLOv11m** at an input resolution of **1280 pixels**. Standard YOLO deployments run at 640px. The reason for doubling it: at 640px, a person standing 15 metres from the camera may occupy only 20-30 pixels of height , reliably missed. At 1280px, they're **detectable**. For campus surveillance where cameras cover large areas, this isn't optional.

Three parameters are tuned beyond YOLO defaults. `classes=[0]` restricts output to persons only , no vehicles, no furniture. `iou=0.70` for non-maximum suppression (default is 0.45) prevents YOLO from collapsing two nearby people into a single merged box. `min_area=400` filters out detections too small to track meaningfully.

The most important decision in the detector is the **dual-confidence split**. Rather than one threshold, detections are partitioned into two groups:

-**High confidence** (conf ≥ 0.40): full tracking, appearance encoding, eligible to create new tracks
-**Low confidence**(0.10 ≤ conf < 0.40): motion-only matching, no encoding, cannot create new tracks

When someone walks behind a pillar, the YOLO's confidence for them drops to around 0.20-0.35. A single threshold at around 0.40 discards them silently, causing the tracker to register missed detections and it eventually leads to killing of that track. The dual split keeps the track alive using **positional proximity** alone, while ensuring sensor noise , which cannot reach 0.40 , never spawns a **ghost track**. This is why tracks survive realistic occlusions rather than dying every time someone walks behind an obstacle.

The ReID Encoder

OSNet_x1_0 is pretrained on MSMT17 , 126,441 images, 4,101 identities, 15 cameras across indoor and outdoor settings. Every confirmed person crop is resized to 128×256 pixels and passed through the network to produce a **512-dimensional embedding** vector. Before storage, the vector is **L2-normalized** :

$$\hat{\mathbf{e}} = \frac{\mathbf{e}}{\|\mathbf{e}\|_2}$$

After normalization, the dot product between any two embeddings is equal to their **cosine similarity** directly. This matters because cosine similarity is the matching operation used everywhere downstream , normalization means no additional division at query time.

Before encoding, **safe crop clipping** handles overlapping detections. When two bounding boxes overlap, each person's crop contains pixels from the other. The system identifies which side the neighbouring person's centre falls on and clips 15% from that edge of the **target crop**. If the clipped result falls below 32×64 pixels, it falls back to the original , a slightly contaminated crop is better than a destroyed one.

Encoding runs only on high-confidence detections. Partially occluded crops produce embeddings that mix two people's appearances, which silently corrupts the **identity gallery**. Restricting encoding to clean, high-confidence crops is a correctness decision, not a speed optimization.

Tracking Module , Kalman Filter and SingleCameraTracker

The Kalman Filter

Each track carries **8-dimensional state vector**:

$$x = [\, cx, cy, w, h, vx, vy, vw, vh \,]^T$$

Position, **size**, and their **velocities**. The velocity terms are what let the filter predict where a person will be before the next frame arrives. When someone is briefly occluded and produces no detection, the Kalman prediction keeps the track positioned correctly so it can be matched when they reappear.

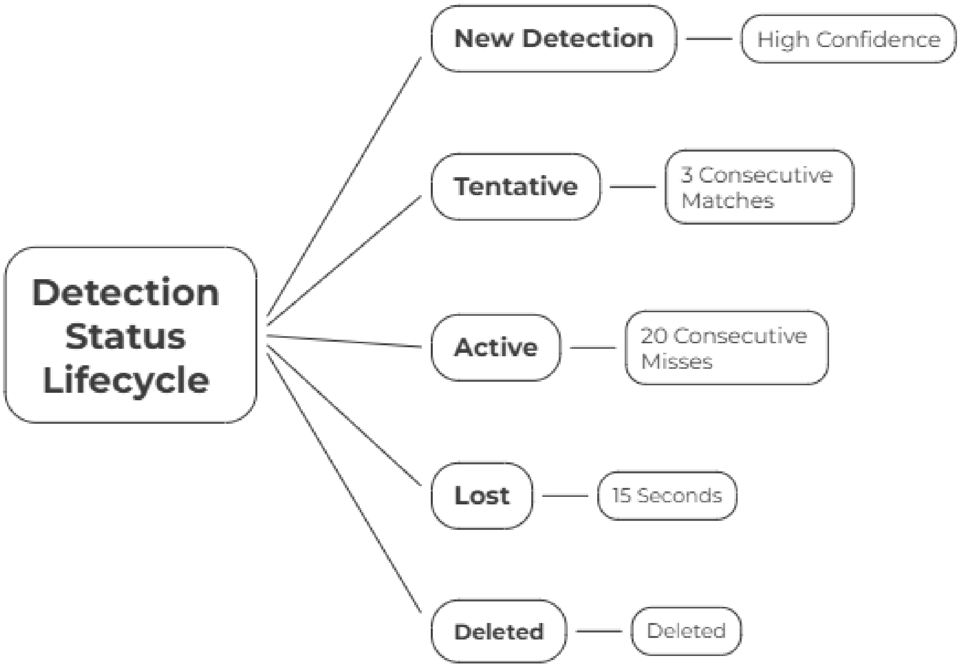
Covariance updates use the **Joseph form** rather than the standard formula:

$$P = (I - KH)\, P\, (I - KH)^T + KRK^T$$

The standard form accumulates floating-point errors over time until the covariance matrix becomes negative-definite , mathematically invalid, and practically means the filter starts producing nonsense. The Joseph form is numerically stable regardless of how many updates run. A security camera tracking a guard post at 30 FPS for eight hours accumulates over 860,000 updates. **Numerical stability** is not a theoretical concern here.

The SingleCameraTracker

Each track lives in one of three states:



TENTATIVE tracks don't appear on the dashboard. They have to survive **three consecutive frames** before being confirmed , this stops momentary noise from ever being displayed as a real person. **LOST tracks** stay alive for **15 seconds**, which is long enough for someone to walk from one camera's coverage zone into another's.

Matching runs as a **four-stage cascade**. Each stage handles a specific combination of track state and detection confidence, and only unresolved tracks and detections move to the next stage. The **cost function** for each (track, detection) pair is:

$$\text{cost} = w_m \cdot C_{motion} + w_a \cdot C_{appearance} + w_i \cdot C_{iou}$$

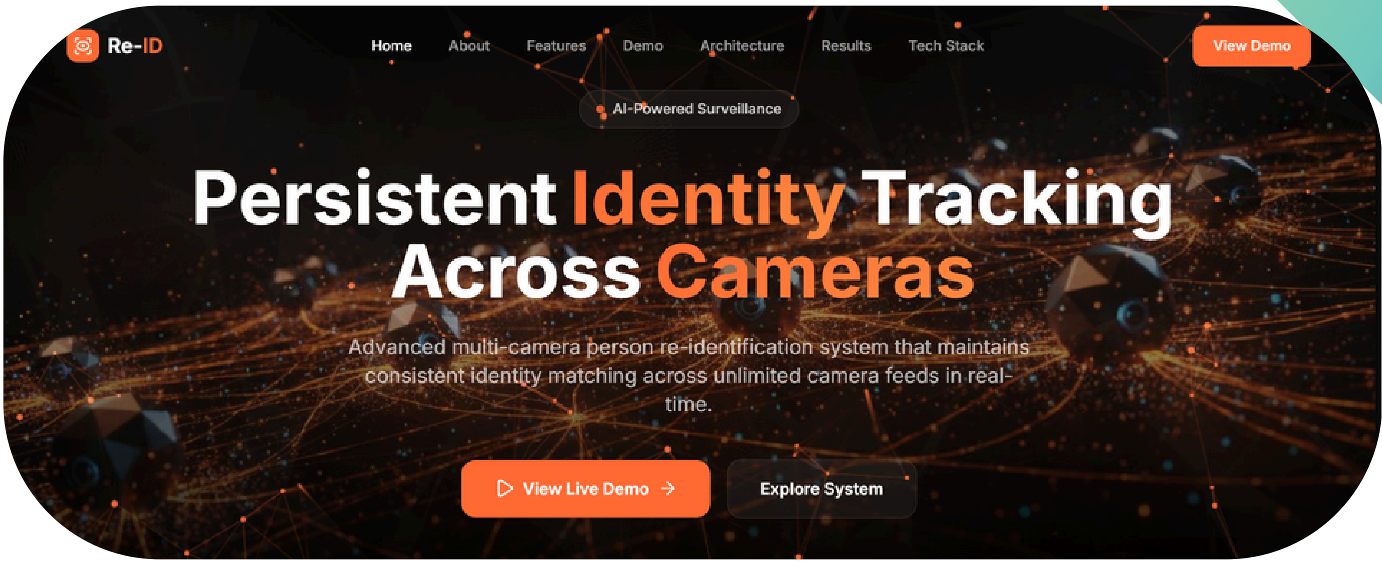
The weights are not global , they're computed **per pair** based on whether the track has a nearby neighbour. **Isolated tracks** get **motion-dominant weights** (0.50, 0.40, 0.10) because Kalman predictions are reliable when no one else is nearby. **Crowded tracks** get **appearance-dominant weights** (0.30, 0.45, 0.25) because when two people stand close together, their predicted positions overlap and motion alone can't distinguish them. This per-pair adaptation is what keeps identity assignments stable in crowded scenes.

Before computing the full cost, a **Manhattan distance gate** runs first:

$$| c_x^{pred} - c_x^{det} | + | c_y^{pred} - c_y^{det} | > d_{(gate)}$$

Pairs which fail this check are imposed a **sentinel cost** of **1e6** and those pairs are skipped completely. In a typical frame this **eliminates around 72%** of **candidate pairs** before any matrix operations run. **Manhattan distance** is used specifically because it needs no multiplications or square roots , just additions and absolute values. Fast enough to run on every pair, every frame.

Frontend , Operator Dashboard



The frontend is a **React** application that connects to the backend over **Socket.IO** on startup and stays connected. No page refreshes, no polling intervals. When the backend emits an event, the dashboard updates within the same second.

The interface is organized around four views:

Live Camera Grid shows **annotated feeds** from all cameras simultaneously. Each detected person has a bounding box, a local track ID, and , once re-identified , their global identity label. The annotations are drawn on frames sent from the backend, not computed in the browser.

Tracking Map shows each **global identity's movement path** across camera zones on a campus floor plan. As re-identification events arrive, the map updates the path in real time. In the context of the hostel theft: this is the view that would show the thief's entry point, route through the building, and exit , all reconstructed live as they moved.

Event Log is a **chronological list** of every system event , person appeared, person re-identified, track lost, alert triggered , with timestamps and identity thumbnails. It's the audit trail. After an incident, this is what security reviews.

Analytics Panel shows **system-level metrics**: active track count per camera, re-identification success rate, detection counts over time. Useful for understanding whether the system is performing well and where coverage gaps might exist.

All four views update from the same **Socket.IO event stream**. There's no separate data pipeline for each view , one connection, all views live simultaneously.

System Working

This Section explain how the components are organized and explains what actually happens when the system runs — what each stage does to the data it receives, what decisions get made, and what comes out the other end. The focus here is the mechanics of one complete processing cycle, from raw frame to dashboard update.

- ## Detection — Finding People in Pixels

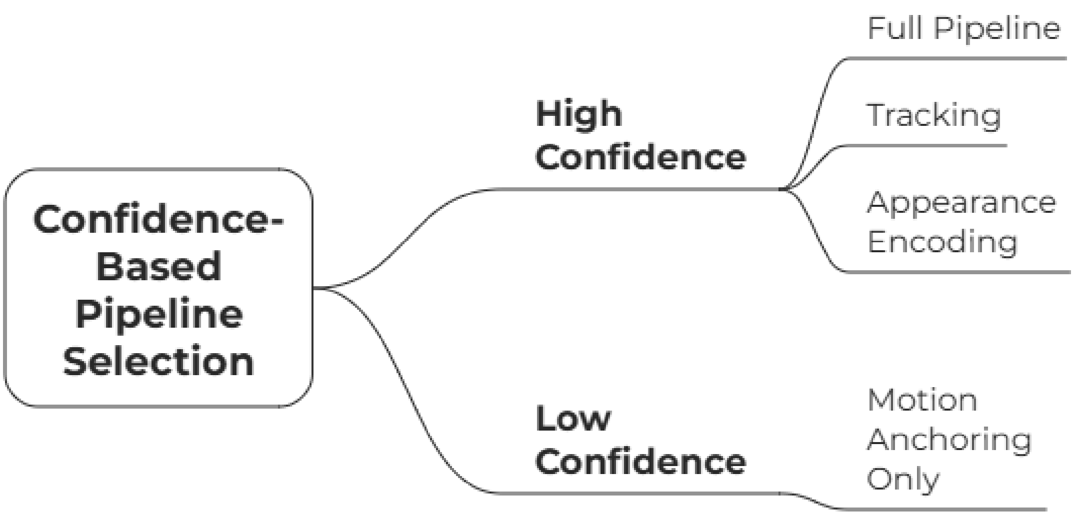
A frame arrives from the camera. It's a grid of pixels. Nothing in it is labelled, nothing is tracked. The detector's job is to find every person in it and return their locations with confidence scores.

YOLOv11m processes the frame at 1280px input resolution. This matters more than it sounds. A person standing 15 metres from a wide-angle corridor camera might occupy 25–30 pixels of height in the frame. At 640px — the standard resolution most deployments use — that person is either missed entirely or detected inconsistently. At 1280px, they appear reliably. The price is compute time per frame, which is a worthwhile trade for a system where a missed detection can break a track.

YOLO returns a list of **bounding boxes**, each with a confidence score. Before anything else happens, three filters run. **Person class only** — any detection that isn't a person is dropped immediately, saving GPU cycles that would otherwise go toward chairs and doorframes. **Minimum area** of **400 square pixels** — detections smaller than this are either background noise, reflections, or people far enough away that tracking them adds nothing useful. And **NMS IoU** at 0.70 rather than the default 0.45 — when two people stand close together, the default setting tends to merge their bounding boxes into one. At 0.70, both survive as separate detections.

What comes out of the detector is not one list. It's two:

Confidence-Based Pipeline Selection



The reason for the split is specific. When someone walks behind a pillar or another person, YOLO's confidence for them drops — they're partially visible, the bounding box is uncertain, maybe only a shoulder or leg is exposed. Confidence lands in the 0.15–0.35 range. A single threshold at 0.40 discards this as noise. The tracker then counts a missed detection, and after enough missed detections, it kills the track. The person re-enters the frame and the system assigns them a new ID. **Cross-camera re-identification** breaks down from that point.

Low-confidence detections fix this. They can't create new tracks — sensor noise also falls in this range and we can't let noise spawn **ghost identities**. But they can match against existing tracks positionally, keeping the track alive through the occlusion without any appearance information. The person emerges from behind the pillar and their original track is still there, waiting.

In **multi-camera mode**, all camera frames are stacked into a single GPU batch call. Four cameras means four frames processed together rather than sequentially — roughly 18ms of GPU kernel-launch overhead eliminated per cycle, independent of what the model actually computes.

- ## Tracking — Turning Boxes Into Persistent Identities

Detection gives positions. Tracking gives continuity. These are different problems.

The tracker runs per-camera. Each camera's **SingleCameraTracker** maintains a set of tracks, each representing one person the system is currently following. Every track carries its current position, velocity, appearance history, and a state label: TENTATIVE, ACTIVE, or LOST.

The first thing that happens each cycle isn't matching — it's prediction. Every track, regardless of state, runs its Kalman filter forward one step. The filter uses the track's estimated velocity to predict where the person should be before any new measurement arrives. This prediction is what makes matching possible: instead of comparing the detector's output against where the person was last seen, the system compares it against where the person is expected to be right now.

Matching then runs as a **four-stage cascade**. ACTIVE tracks get first claim on high-confidence detections. The reason for the priority ordering is that established tracks with confirmed appearance histories should get matched before tentative ones. Without this ordering, a newly appearing person's detection can be claimed by an older track through positional proximity alone, swapping identities in a way that's hard to debug.
For each surviving (track, detection) pair, a **blended cost** is computed:

$$\mathbf{cost} = w_m \cdot C_{motion} + w_a \cdot C_{appearance} + w_i \cdot C_{iou}$$

The weights shift depending on the local geometry. If another track's predicted position overlaps this track's (IoU > 0.15), that track is in a **crowded region** — motion becomes an unreliable discriminator because two people standing close together have similar predicted positions. Appearance gets higher weight. If the track is isolated, motion is reliable and takes priority. This check runs once per track per frame and gets looked up during cost computation — $O(T^2)$ **precomputation**, $O(1)$ **per pair**.

Before computing the full blended cost, a **Manhattan gate** eliminates most pairs.

$$|c_x^{pred} - c_x^{det}| + |c_y^{pred} - c_y^{det}| > d_{(gate)} \Rightarrow \text{skip}$$

Around **72%** of candidate pairs fail this check. That's about **75,000** floating-point operations per frame that never run. **Manhattan** distance needs only additions and absolute values — no **square roots**, no **multiplications**. It's cheap enough to run on every pair before anything expensive starts.

Pairs that pass the **Manhattan gate** face a **Mahalanobis** hard gate at chi-squared threshold **9.48** (95th percentile, **4 degrees** of freedom). This rejects pairs that are **spatially** close but statistically unlikely given the track's current **uncertainty estimate.** Only what survives both gates enters the final **Hungarian** assignment.

After matching, unmatched tracks face a specific check before their miss counter increments. If an unmatched track's predicted position overlaps a matched active track with **IoU > 0.30**, it's classified as occluded — hidden behind the person who just got matched. Its miss **counter** doesn't increment. It holds its position at the **Kalman prediction**. This is the **mechanism** that lets tracks survive one person walking behind another without dying.
Tracks that accumulate **20 consecutive** missed frames move to the lost buffer, where they remain for **15** seconds. If a person reappears — either on the same camera or a different one — within that window, the track can be revived. After **15** seconds, it's deleted and a **TRACK_ENDED** event fires.

New detections that weren't claimed by any existing track go through birth guards before a new track is created: aspect ratio check (**width/height > 1.5** means **YOLO** probably merged two people into one box), **spatial duplicate** check against existing tracks, and appearance similarity against the lost buffer. Only **detections** that clear all guards become TENTATIVE tracks. They need **3 consecutive** confirmed matches before becoming **ACTIVE** and appearing on the dashboard.

- ## Re-ID — Connecting Identities Across Cameras

At this point the system knows who is where on each camera independently. Camera 1 has **Track #3** and **Track #5**. Camera 2 has **Track #7** and **Track #9**. These are four separate local identities with no connection to each other. **Re-identification** creates that connection.

Every ACTIVE track gets its appearance encoded by **OSNet_x1_0**. The person's **bounding box** is cropped from the frame, resized to **128×256 pixels** — tall and narrow, the standard aspect ratio for person **ReID** — and passed through the network to produce a 512-dimensional embedding vector. The vector is **L2-normalized** before storage, so dot products equal **cosine similarities** directly.

Before encoding runs, **overlapping** crops are clipped. When two bounding **boxes overlap,** cropping directly produces an image that contains pixels from both people. The resulting embedding is a blend of two appearances — it will match neither person **reliably** and will corrupt whichever track it gets pooled into. For each overlap, the system identifies which side the other person's centre falls on and clips **15**% from that **edge**. If the clipped result is smaller than **32×64 pixels**, it falls back to the original — a slightly contaminated crop is better than one that's mostly **background**.

Each track maintains a pool of its **top-5 highest**-confidence embeddings using a **min-heap**. New embeddings replace the lowest-confidence entry in the heap only if they score higher. The identity vector used for matching is the median across the **pool's five** embeddings — not the mean. If one of the five crops was taken during partial occlusion and produces a corrupted embedding, the mean gets pulled toward it. The **median ignores** it.

The **GlobalMatcher** holds a gallery of all known identities from all cameras. When a new ACTIVE track appears — either freshly created or crossing into a new camera's view — its embedding is compared against the gallery:

$$\mathbf{s = Egallery \cdot \hat{e}query \in \mathbb{R}^N}$$

This is implemented as a single matrix multiplication against the pre-built **(N × 512)** gallery **matrix**, so all N similarity scores are computed in one **BLAS** call. The gallery matrix is rebuilt only when entries change, not every frame, which keeps the system **efficient**

A match is accepted only if the top similarity clears the gallery **threshold (0.60)** and beats the **second-best** match by at least **0.08.** The margin gate is what prevents **ambiguous** matches from being forced.

If a person's **embedding** is genuinely halfway between two gallery entries — say, they've changed **lighting** conditions drastically or turned around — the system abstains rather than guessing. The track gets a new **global** ID and the ambiguity is resolved as more **embeddings** accumulate.

A confirmed match triggers a **PERSON_REIDENTIFIED** event. The new track inherits the global ID of its **gallery match**. From this point, both the local track on the new camera and the existing global identity in the gallery are **linked**.

Event Generation — Discrete Signals From Continuous Data

The system does not transmit per-frame tracking data to the frontend, as this would be noisy, bandwidth-intensive, and difficult for operators to interpret. Instead, it follows an **event-driven approach**, where the pipeline generates discrete, meaningful events, each carrying sufficient context for **actionable decision-making**.

Two key event types illustrate this design. The **TRACK_ACTIVATED** event is generated when a tentative track achieves sufficient confidence through consecutive matches and is promoted to an active state, indicating the **confirmed presence** of an individual in the scene. The **PERSON_REIDENTIFIED** event occurs when the system successfully links a newly detected track to an existing identity across cameras, signifying **cross-camera movement**. This event enables the system to update the tracking map by extending the individual’s movement path across different zones.

TRACK_LOST— 20 consecutive frames with no **matching** detection. The person has disappeared from this **camera's** view. The track moves to the lost buffer.

TRACK_ENDED — a lost track has expired from the 15-second buffer without **revival**. The person has left the **monitored** area or moved somewhere no camera covers. The gallery entry is **retired**.

Each event carries a timestamp, camera ID, local track ID, global identity **ID** where applicable, and a **thumbnail** crop of the person at the moment of the event. The **Orchestrator** collects all events from all cameras each cycle and routes them over **Socket.IO** to the **frontend**. From event **occurrence** to dashboard update, the latency target is under one second.

The event stream also drives alerts. Specific patterns — a **PERSON_REIDENTIFIED** for a flagged identity, a **TRACK_ACTIVATED** in a designated restricted zone, a TRACK_LOST with no downstream camera pickup within the expected **transit** window — trigger notifications to security **personnel**. The alert carries the same context as the **underlying** event: who, where, when, and a visual.

Visualization — Making the Data Readable

The dashboard receives everything over the same Socket.IO connection the events use. There's no separate **data pipeline**, no database read on the frontend side. When an event arrives, the relevant view updates. All four views stay live simultaneously from one connection.

Live **Camera** Grid receives annotated **JPEG** frames from the backend each cycle. Bounding boxes and identity labels are drawn server-side before the frame is transmitted. The **frontend** renders what it receives — no computer vision in the browser. This keeps the client **lightweight** and ensures every **operator** sees exactly the same annotations regardless of their device.

Tracking Map updates on PERSON_REIDENTIFIED events. Each global identity has a path drawn on the campus floor plan. As the **identity moves** from camera to camera, a new path segment is added connecting the previous zone to the current one.

A person who entered at the main gate, appeared in the corridor camera, and then showed up near the hostel block has a **visible connected** path across all **three zones** — built live as the events arrive, not reconstructed after the fact.

Event Log is a scrolling **timeline** of every system event — activations, re-**identifications**, losses, expirations, alerts — with timestamps and thumbnails. In normal **operation**, security can glance at it to see what the system has seen. After an incident, it's the audit trail. Every identity transition through the camera network is logged with enough timestamp precision to **reconstruct** a minute-by-minute account of where someone was and when.

Analytics Panel shows per-camera active track counts, re-identification rates over recent time windows, and system health metrics. A sudden drop in active tracks on one **camera** without corresponding **re-identifications** on adjacent cameras is worth **investigating** — it might indicate a camera fault, a **lighting** change that's degrading detection, or someone moving through a coverage gap in the network.

The entire interface is **operationally** passive from the security officer's **perspective**. They don't configure anything during **normal** operation. The system runs, the **dashboard** updates, and alerts fire when they need to. The officer's job is to respond to what the system surfaces, not to manage the system itself.

Novelty of the Proposed System

CROSS-CAMERA IDENTITY CONTINUITY

Most surveillance systems developed at the academic level focus on detecting individuals and assigning temporary identities within a **single camera**. While such systems may perform well in controlled environments, they often fail in **real-world scenarios** where individuals move across **multiple cameras**. The novelty of the proposed system lies in the integration of multiple components that enable **reliable and consistent performance** in practical settings.

A key contribution of this system is **cross-camera identity continuity**, which allows the system to recognize that a person appearing in one camera is the **same individual** observed earlier in another camera. This is achieved **without** the use of **GPS**, **wearable devices**, or any active tagging, relying purely on **visual appearance**.

To enable this in real-world conditions, the system uses a **lost buffer** to preserve identities during transitions and a **two-stage matching process** to ensure **accurate identity association** while avoiding incorrect matches. Together, these components ensure **stable identity tracking** even in the presence of **occlusions** and **camera transitions**.

Engineering Innovations

ENGINEERING-LEVEL INNOVATIONS

STANDARD APPROACH	PROPOSED SYSTEM	IMPACT
Single confidence threshold	Dual-confidence handling	Maintains tracking during occlusion
Single-stage assignment	Multi-stage matching cascade	Preserves identity stability
Uniform crowd handling	Adaptive matching	Improves accuracy in crowded scenes
Simple track termination	Occlusion-aware tracking	Prevents identity loss
Fixed thresholds	Adaptive thresholds	Improves re-identification reliability

These improvements enhance the system's ability to operate reliably in **real-world conditions**. By addressing challenges such as occlusion, **crowding**, and **identity ambiguity**, the proposed approach ensures more **stable** and **accurate tracking** compared to conventional methods.

System Optimisations

The proposed system **integrates** a series of low-level algorithmic and data-structure **optimisations** to ensure efficient execution in a real-time multi-camera tracking environment. These **optimisations** are designed to reduce **computational** overhead while **preserving** identity consistency across frames and camera transitions.

MANHATTAN DISTANCE PRE-GATE

A lightweight L1 distance filter is used to eliminate spatially unlikely **track–detection** pairs before expensive matching. This early pruning significantly reduces the number of candidate **associations** processed per frame.

$$d_{L1}(x, y) = |x_1 - y_1| + |x_2 - y_2|$$

MIN-HEAP BASED EMBEDDING POOL

Each track maintains a **bounded set** of high-confidence embeddings using a **min-heap structure**, ensuring that only reliable **identity** features are retained. A **median**-based representation reduces the effect of noisy or corrupted samples.

$$\text{Insertion Complexity} = O(\log k), \quad k = 5$$

VECTORIZED GALLERY MATCHING

Identity matching is performed using vectorized matrix operations instead of **sequential comparisons**, enabling parallel similarity computation and **significantly** improving runtime **efficiency**.

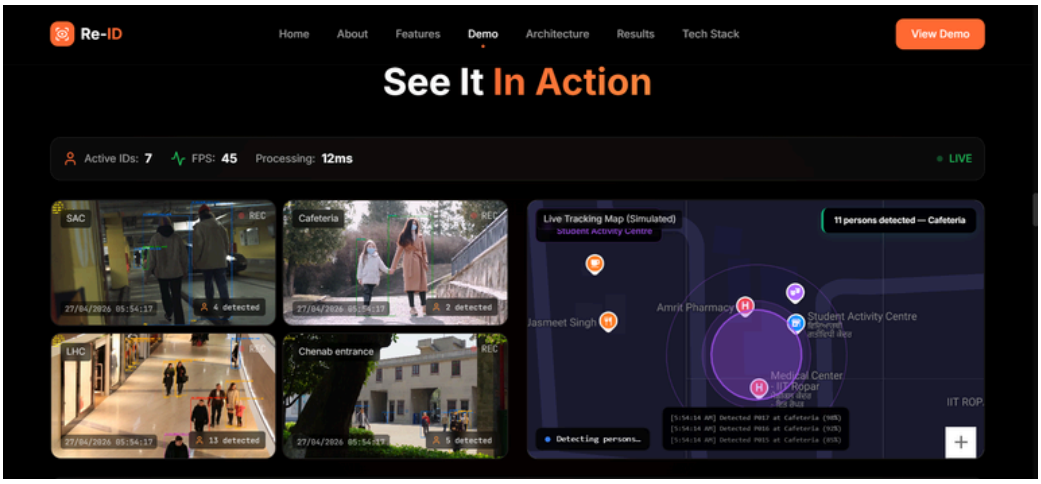
$$O(1) operations$$

MULTI-STAGE CASCADE ASSIGNMENT

The matching process is **decomposed** into multiple stages, where high-confidence **associations** are resolved first and **ambiguous** cases are handled later. This reduces identity **switches** and improves stability.

$$O(max(T, D)3)$$

Real-Time Alert and Integration Architecture



The system follows an event-driven architecture, where meaningful events such as **identity detection**, cross-camera transitions, and track termination trigger **immediate alerts.** Instead of relying on continuous manual monitoring, the system **actively** identifies important **situations** and responds in real time.

When an event is generated, the central orchestrator emits a structured signal through a **persistent communication channel**. Each alert contains contextual information including the detected identity, timestamp, camera source, and a **visual snapshot**. This ensures that security personnel receive actionable information instantly **without needing** to interpret raw video feeds.

The **frontend** maintains a **persistent connection**, allowing all system components to remain synchronized in **real time**. A single connection **continuously** streams updates across multiple views, including live feeds, identity tracking, and **event logs.** This eliminates the need for polling, refresh cycles, or separate data pipelines.

Unlike typical academic implementations that rely on pre-recorded inputs or **offline processing**, the proposed system operates as a fully integrated pipeline where cameras, **tracking models,** and user interfaces **function together** in real time. The system is no simulation of an idea, but a working implementation **depl** actual hardware.

Large-Scale Implementation

The AI-based **multi-camera** tracking and **re-identification** system, although initially built for a hostel setup, can easily be expanded to much **larger environments**. Since the system is designed for real-time **detection**, **tracking**, and **maintaining** identity across cameras, it can be adapted for campuses, public spaces, and even city-level **surveillance**. However, scaling it up requires careful planning in terms of infrastructure, computing power, and **system integration**.

Deployment in Institutions

At the institutional level, this system can go beyond a single hostel and cover the entire campus—**academic buildings**, hostels, **admin blocks**, entry gates, parking areas, and more.
Instead of **monitoring cameras** separately, everything can be connected to a central system. This gives security teams a complete picture of movement across campus. They can track where a person came from, where they went, and identify unusual **patterns** more easily.
It's especially useful in **sensitive** areas like entry gates or **restricted** zones. The system can actively monitor activity and send alerts when something **suspicious** happens. In short, it turns **surveillance** from passive recording into an intelligent, real-time support system.

Use in Smart Cities and Public Spaces

This system can also be applied to large public spaces like railway stations, airports, malls, and city streets.

For example:

- It can help track **missing persons** in crowded places
- Detect suspicious behavior in real time
- Assist during **emergencies** like evacuations

In a smart city setup, this system can integrate with other data sources like **traffic systems** and **public safety networks.** This creates a connected surveillance ecosystem where decisions are based on real-time data rather than just recordings.

Technical Scalability

Scaling the system comes with technical challenges, especially when **handling many cameras.**

- **Distributed processing:** Instead of **one server, multiple GPU servers handle** different camera feeds. This keeps the system fast and efficient.
- **Cloud integration:** Data like video feeds, logs, and tracking info can be **stored** and **processed in the cloud**, allowing flexibility and scalability.
- **Network requirements:** High-speed and reliable internet is essential to ensure **smooth real-time data transfer.**

The system already uses optimizations like batched GPU processing and modular design, which makes scaling more manageable.

Working with Existing CCTV Systems

A key advantage of the proposed system is that it **does not require replacing existing CCTV infrastructure**. Instead, **it can integrate directly with current camera feeds**, such as RTSP streams, and operate as an additional AI layer. This significantly **reduces deployment cost** and simplifies implementation.

The system also supports a **centralized dashboard** that enables security personnel to monitor live feeds, track individuals, and access **event logs and alerts** in real time. This integrated interface improves operational efficiency and makes the monitoring process more structured and user-friendly.

Constraints

Even though the system is powerful, there are **several** real-world challenges that need to be considered.

Hardware Limitations

Running AI models in real time requires **strong hardware,** especially **GPUs**. As the number of **cameras increases**, so does the computational load.
In places with limited resources, maintaining such **infrastructure** can be **difficult**, which may affect performance.

Cost Challenges

Large-scale deployment of surveillance systems involves **significant cost considerations**. Expenses arise from installing additional cameras, procuring **servers and GPUs** for processing, and establishing reliable **network infrastructure**. In addition, **ongoing maintenance** and operational costs must be accounted for over time. These factors can make implementation challenging, particularly for **smaller organizations with limited resources**..

Network and Infrastructure Issues

The system is highly dependent on **stable internet connectivity** and a **reliable power supply** for effective operation. In practical scenarios, issues such as **network delays**, **bandwidth limitations**, and power outages can disrupt real-time monitoring and **impact overall system reliability**.

Privacy and Ethical Concerns

Tracking individuals across locations raises important **privacy concerns**, particularly if data is not managed responsibly. There is a **risk of misuse** in the absence of proper safeguards, making it essential to implement **strict access control**, **secure data storage**, and **clearly defined usage policies**. Additionally, maintaining transparency in how data is collected and used is crucial. Achieving an effective **balance between security and privacy** remains a key consideration in the deployment of such systems.

Accuracy in Real-World Conditions

System **performance can degrade** under challenging conditions such as **poor lighting**, **crowded environments**, and the presence of individuals with **similar appearances**. These factors can lead to **tracking inaccuracies** and identity confusion, highlighting the need for **continuous model improvement** and system refinement.

System Management at Scale

As the **system scales**, its management becomes increasingly complex. Challenges include handling **large volumes of data**, ensuring **synchronization across multiple components**, maintaining **real-time performance**, and performing regular updates and monitoring. These factors highlight the need for **automation and efficient management** tools to ensure smooth and reliable system operation.

Conclusion

Several laptops was stolen in the Chenab Hostel. Cameras were installed. No perpetrator was identified.

It all began here , not through a theoretical paper, but a **case** that had taken place on campus and brought to light the existence of an issue for which there were no **satisfactory answers.** The video feed was there. But **not a mechanism** to link camera to camera, to **track individuals** around the building, or to notify security personnel before time had slipped by.

The Issue

The current state of surveillance technology on campuses is **passive.** Cameras record and guards view footage only during post-incidence analysis. This was evident from the incident at the hostel as well as from the findings of our field research. It also stood out clearly in our **stakeholders' interviews** with both DRDO scientists and campus guards. Our survey, with 1,800+ respondents, further **revealed** that although the presence of cameras was generally acknowledged, their **ineffectiveness was the norm.**

The Solution

Our solution addresses a fundamental question: given a network of camera feeds, how can individuals be **consistently identified across all frames**, and how can the system determine whether a person detected in one camera corresponds to someone previously observed in another?

The pipeline begins with person detection using YOLOv11m, enhanced with a **dual-confidence strategy** that allows identities to be **maintained even during partial occlusions**. This ensures continuity in tracking despite temporary drops in detection confidence.

A **Kalman filter** is used to predict object positions across frames, ensuring continuity during temporary detection gaps. The system employs a **multi-stage cascade tracker** for detection-to-track assignment, which prioritizes stable tracks and prevents new detections from disrupting existing identities. Each detected individual is encoded into a **512-dimensional feature vector** using OSNet for robust representation.

Cross-camera matching is performed using a **vectorized similarity search**, augmented with margin-based filtering and transit-time validation to improve accuracy. The system operates in an **event-driven** manner, with updates transmitted over Socket.IO within one second, enabling the dashboard to reflect changes in **real time**.

The Solution

The immediate impact of the proposed system is the ability to address security incidents in **real time**, **rather than relying on post-facto analysis**. More importantly, it fundamentally changes the nature of surveillance by making monitoring **independent of continuous human attention**, ensuring that critical events are not missed due to manual oversight.

The architecture is not limited to a single hostel and can **scale to larger environments** such as railway stations, public venues, and institutional campuses—anywhere a network of cameras exists but **lacks intelligent coordination**. While CCTV infrastructure is already widely deployed, the real gap lies in the **absence of intelligence** to interpret and connect information across feeds. This system aims to bridge that gap.

The incident at Chenab Hostel highlights this issue clearly. The failure was not due to a lack of cameras, but due to the **absence of an intelligent system** capable of analyzing and correlating the available data in real time.